Unified Study Definitions Model

All Your Questions Answered ☺

12th June 2025

Dave Iberson-Hurst

Disclosure

- I am currently working on contract to CDISC as the CDISC USDM Product Owner
- The views expressed during this presentation are my own and **NOT** those of CDISC
- You will see many CDISC slides. They were all created by myself, except for the 360i slide. All have been presented at various public events.
- There is a Transcelerate slide within the presentation. This has been presented within public forums.

Questions

Questions People Ask

- What is it?
- ICH M11?
- How do the pieces fit together?
- BCs Downstream …?
- Benefits?
- What's Next?

Photo by Anna Shvets, via pexels

What is it?

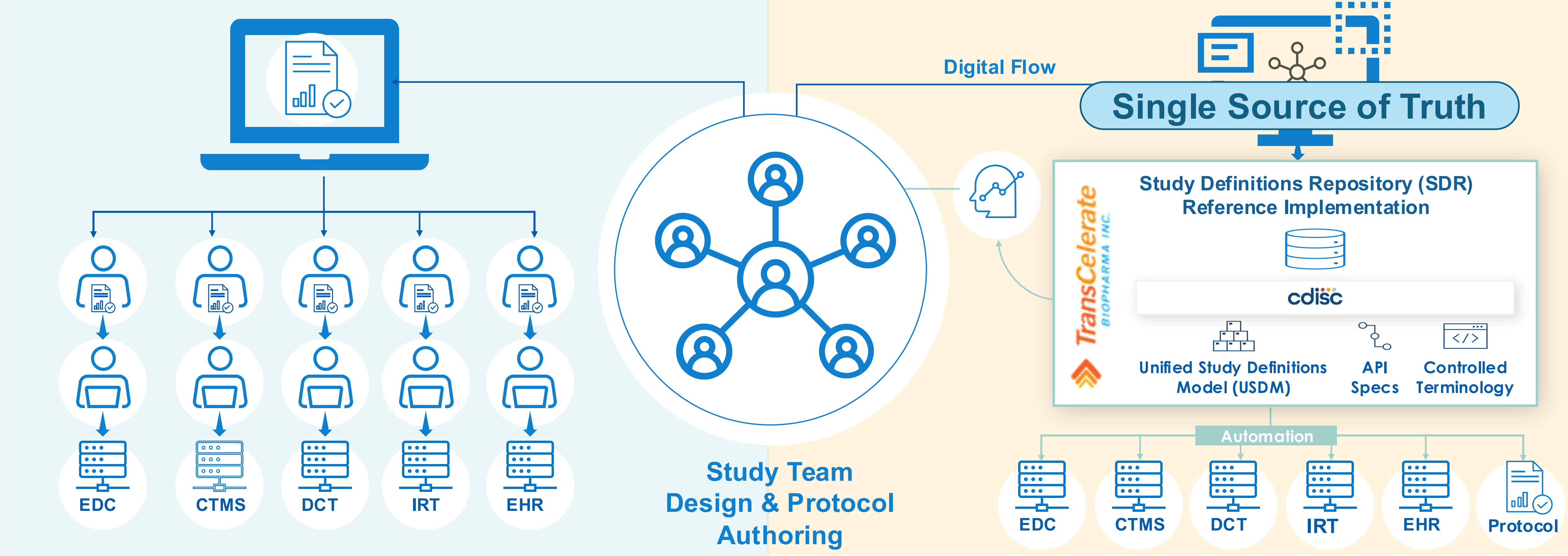
https://www.transceleratebiopharmainc.com/assets/digital-data-flow-solutions/

TransCelerate Digital Data Flow (DDF) Ambition

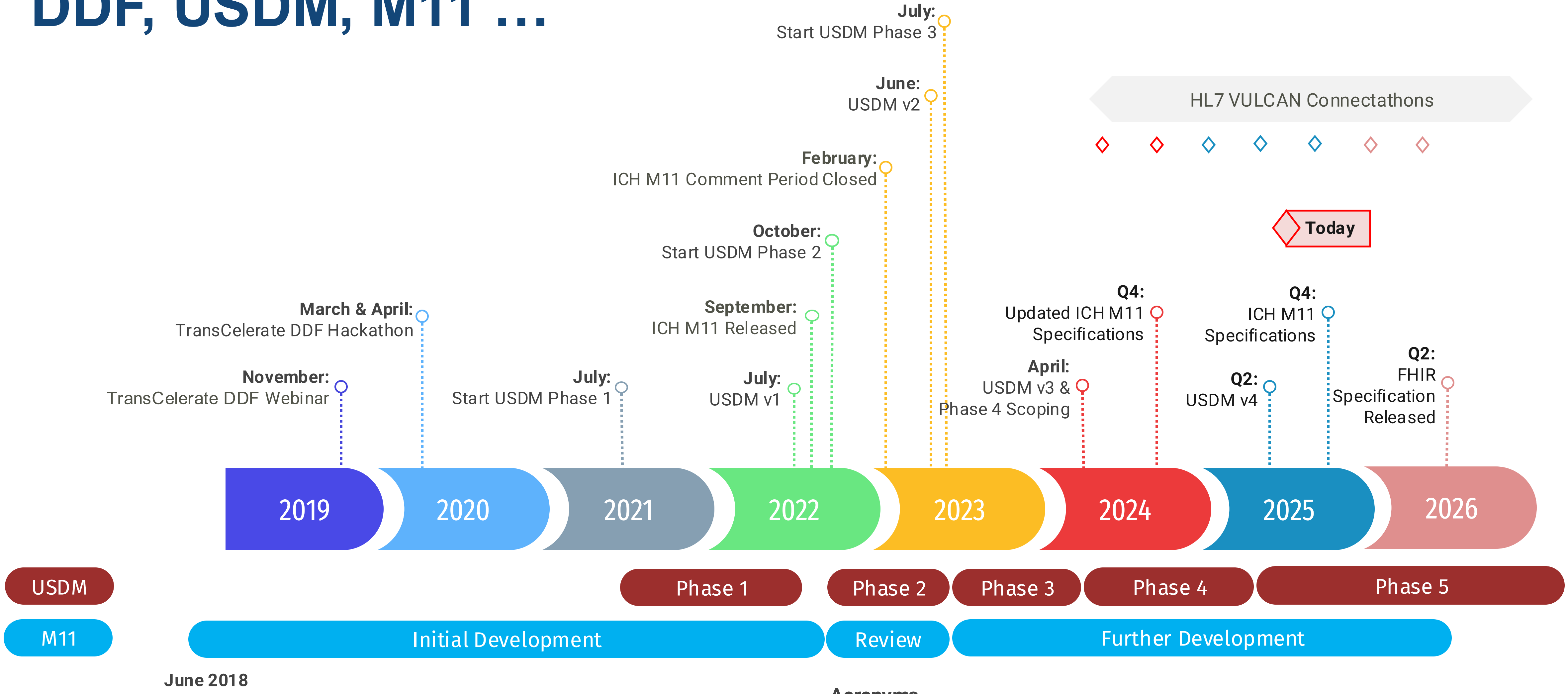
Write Once, Read Many

TODAY: Document-based paradigm for protocol creation, interpretation, and transcription into consuming systems

TOMORROW: Digital paradigm for protocol creation, with fully automated data flow and interoperability between systems



DDF, USDM, M11 …

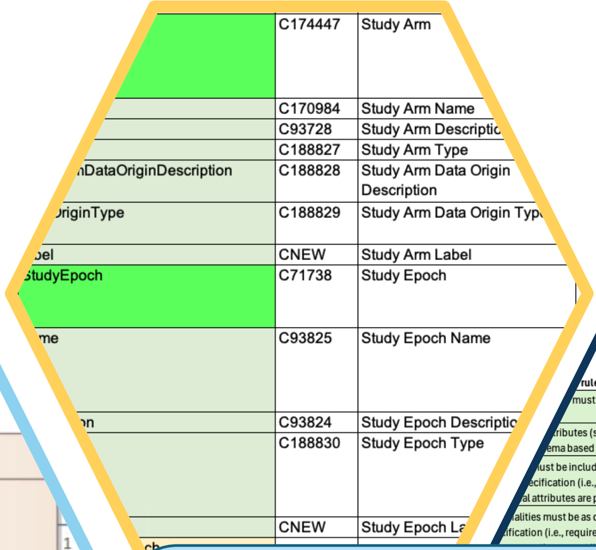


Acronyms
DDF: Digital Data Flow
USDM: Unified Study Definitions Model
ICH: International Council for Harmonisation
M11: Clinical electronic Structured Harmonised Protocol (CeSHarP)

The USDM Standard

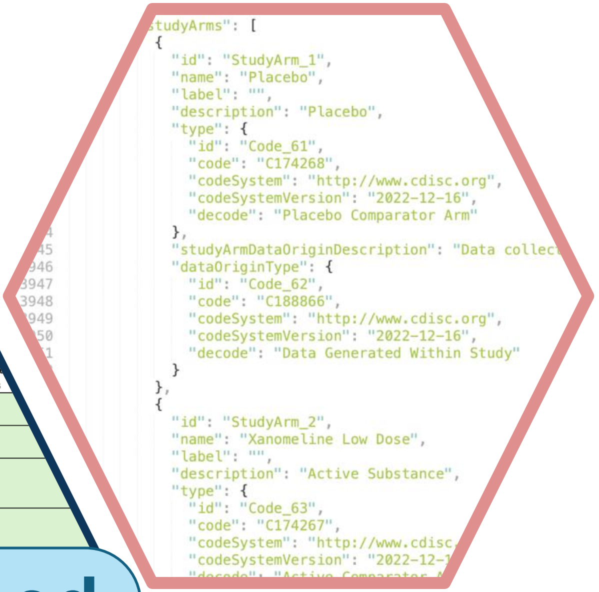
CDISC Controlled Terminology

Provides further semantics, complementing the UML model. Includes the definition of classes, attributes, and value sets



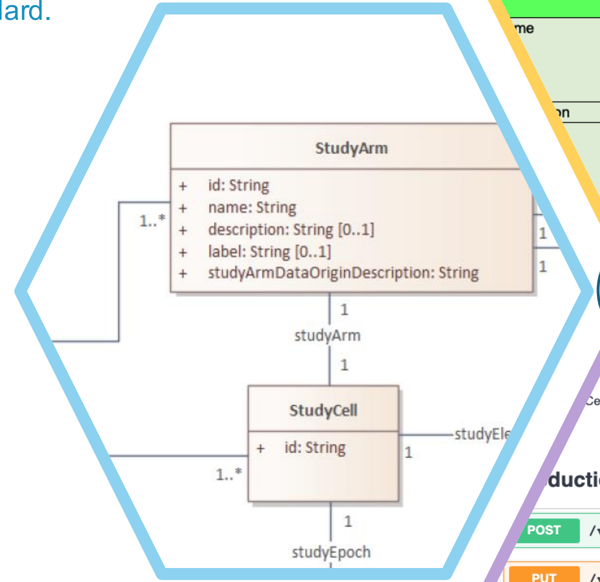
Examples

Example protocols implemented in the USDM with associated JSON files and visualisations



Logical Model

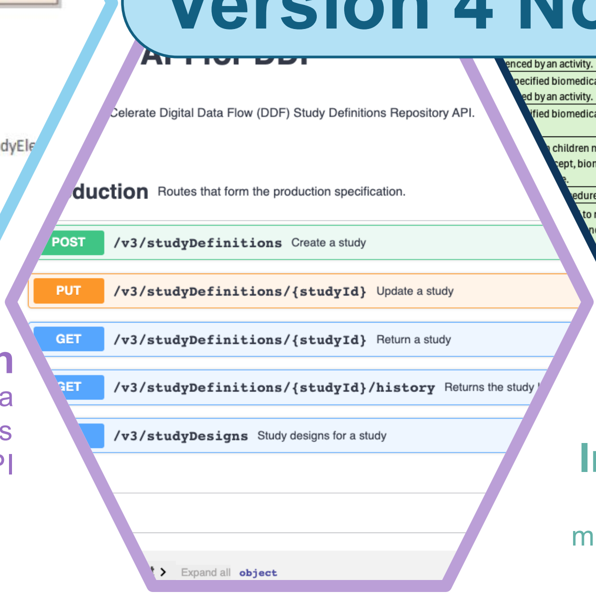
The UML logical model (a class diagram) that provides the basis for the USDM standard.



CORE Rules

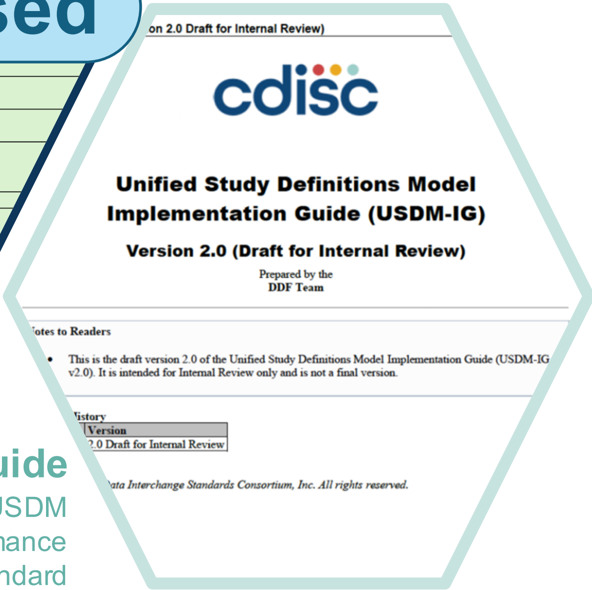
Specification of the rules that define USDM compliance

Version 4 Now Released



API Specification

Provides the means to exchange a single study between machines using a JSON API

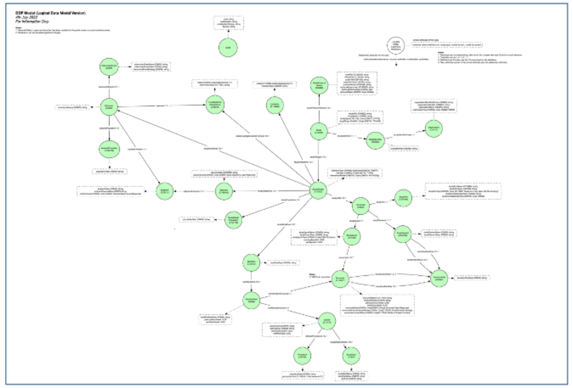


Implementation Guide

Guidance on using the USDM model and ensuring conformance with the standard

CDISC DDF / USDM, Phases One to Four

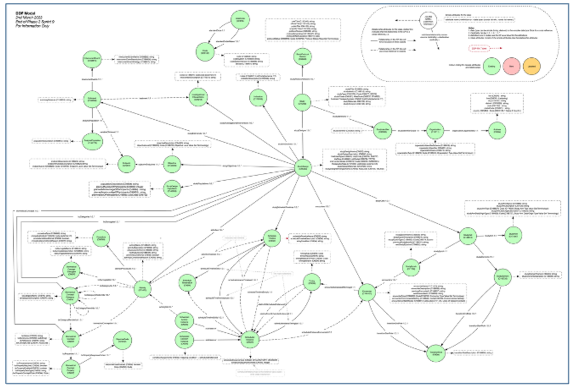
Phase One



25 Classes

- Solid foundation
- The protocol document was an external entity into which the structured content could be exported

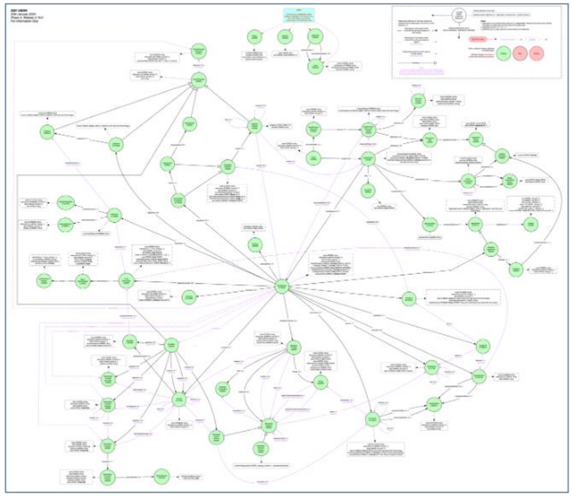
Phase Two



35 Classes

- Focused on the structured elements of the protocol, e.g. the Schedule of Activities (SoA) & Biomedical Concepts (BCs)
- The protocol document still an external entity

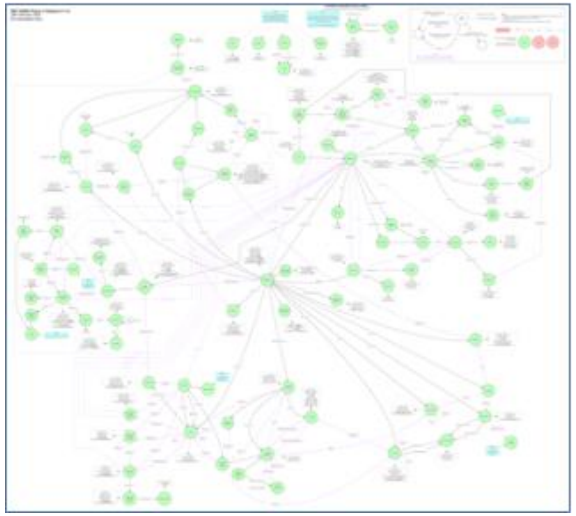
Phase Three



58 Classes

- Now contains structured and unstructured elements
- The entire protocol document can be held within the USDM
- Allows for the protocol document to be generated from the model

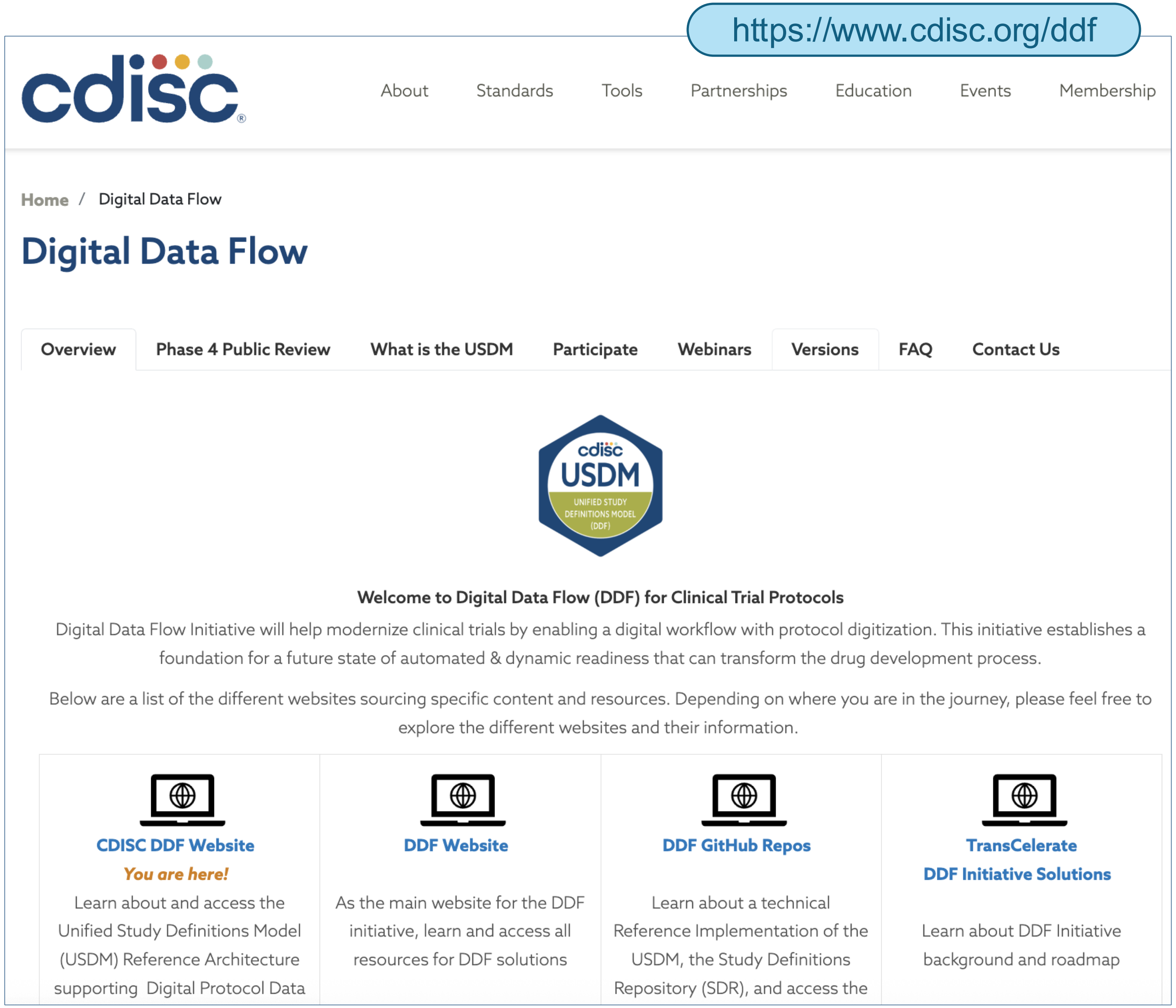
Phase Four



89 Classes

- Aligned with ICH M11
- Support for complex studies, interventional & observational studies, and medical devices
- Maximise content re-use and support for multiple document templates
- Extension mechanism to provide flexibility

DDF Web Page



https://www.cdisc.org/ddf

About | Standards | Tools | Partnerships | Education | Events | Membership

Home / Digital Data Flow

Digital Data Flow

Overview | Phase 4 Public Review | What is the USDM | Participate | Webinars | Versions | FAQ | Contact Us

Welcome to Digital Data Flow (DDF) for Clinical Trial Protocols

Digital Data Flow Initiative will help modernize clinical trials by enabling a digital workflow with protocol digitization. This initiative establishes a foundation for a future state of automated & dynamic readiness that can transform the drug development process.

Below are a list of the different websites sourcing specific content and resources. Depending on where you are in the journey, please feel free to explore the different websites and their information.

CDISC DDF Website
You are here!
Learn about and access the Unified Study Definitions Model (USDM) Reference Architecture supporting Digital Protocol Data

DDF Website
As the main website for the DDF initiative, learn and access all resources for DDF solutions

DDF GitHub Repos
Learn about a technical Reference Implementation of the USDM, the Study Definitions Repository (SDR), and access the

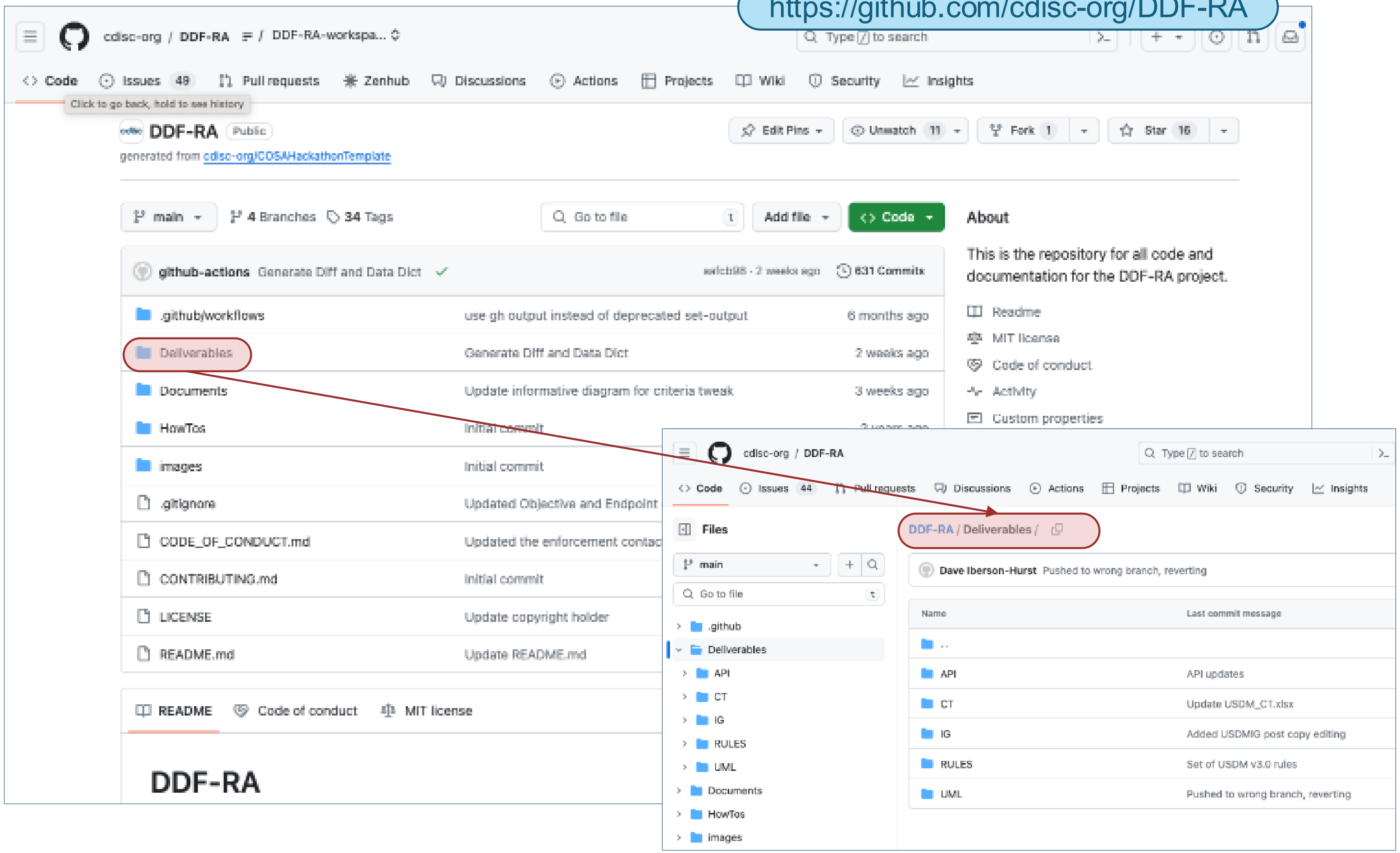
TransCelerate DDF Initiative Solutions
Learn about DDF Initiative background and roadmap

CDISC DDF

Scan Me

DDF-RA (USDM) GitHub

https://github.com/cdisc-org/DDF-RA



CDISC GitHUb

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ICH M11?

M11 Is …

ICH CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL (CeSHarP)

https://www.ich.org/page/multidisciplinary-guidelines

ICH harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL (CESHARP)

M11

Draft version

Endorsed on 27 September 2022

Currently under public consultation

At Step 2 of the ICH Process, a consensus draft text or guideline, agreed by the appropriate ICH Expert Working Group, is transmitted by the ICH Assembly to the regulatory authorities of the ICH regions for internal and external consultation, according to national or regional procedures.

Provides background, purpose, and scope as a guideline

ICH harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL (CESHARP)

M11 TEMPLATE

Draft version

Endorsed on 27 September 2022

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Provides the written format for the Interventional Clinical Trial Protocol Template

ICH harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

CLINICAL ELECTRONIC STRUCTURED HARMONISED PROTOCOL (CESHARP)

M11 TECHNICAL SPECIFICATION

Draft version

Endorsed on 27 September 2022

Currently under public consultation

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Provides the technical representation aligned with the guideline and protocol template

ICH M11 WG

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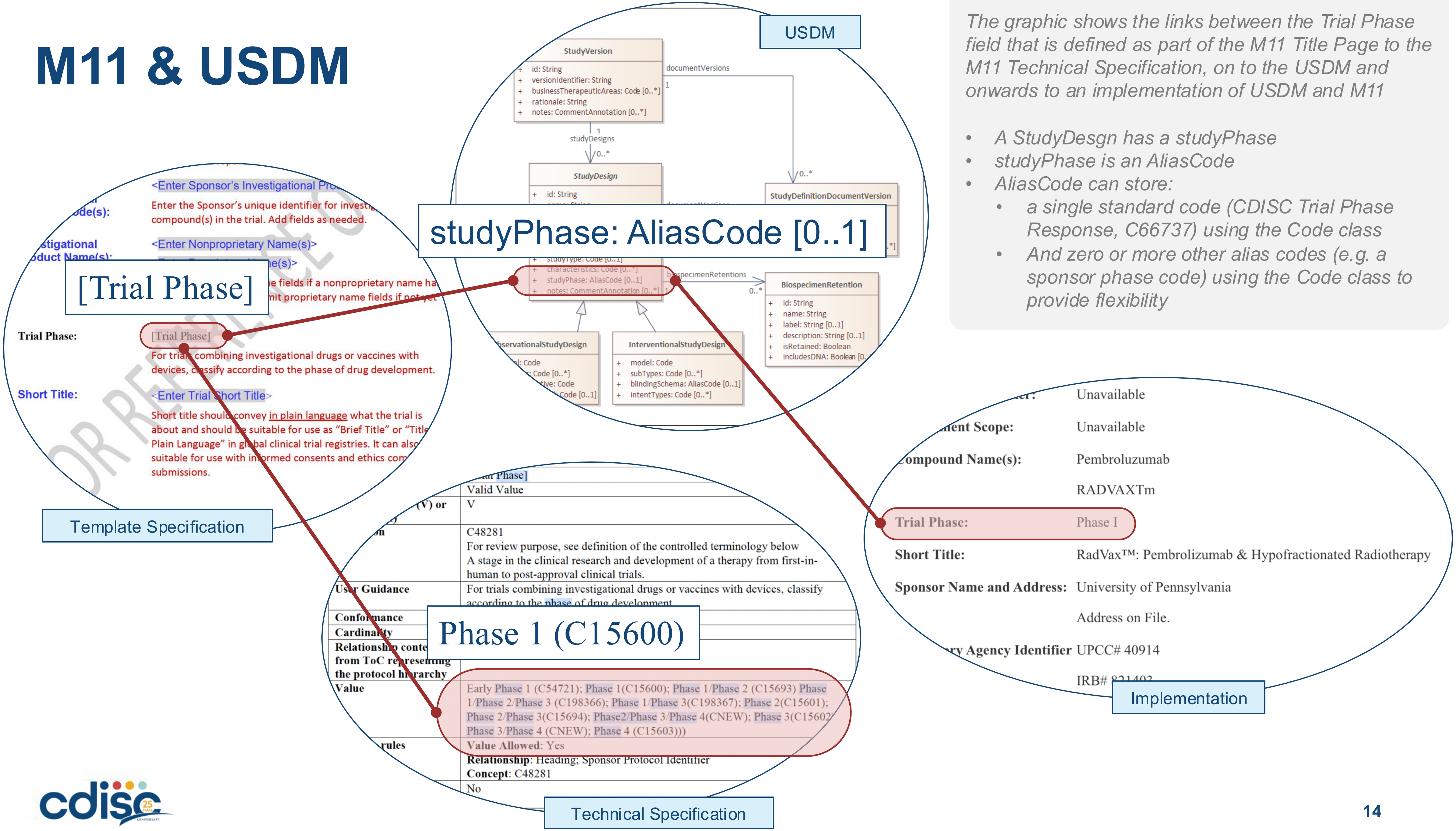
M11 & USDM



The graphic shows the links between the Trial Phase field that is defined as part of the M11 Title Page to the M11 Technical Specification, on to the USDM and onwards to an implementation of USDM and M11

- *A StudyDesgn has a studyPhase*
- *studyPhase is an AliasCode*
- *AliasCode can store:*
  - *a single standard code (CDISC Trial Phase Response, C66737) using the Code class*
  - *And zero or more other alias codes (e.g. a sponsor phase code) using the Code class to provide flexibility*

ICH and CDISC MOU (Memorandum of Understanding)

As a collaboration between ICH and CDISC, the goals of the agreement are to:

- Use a unified governance process and terminology services for the long-term support of ICH controlled terminologies
- Curate and maintain ICH controlled terminologies
- Follow a robust process for the public review and publication of ICH terminologies
- Ensure the terminologies are freely available to the public following public review

Scope

For ICH members to adopt and implement a clinical information standard it is critical that all terminology components, including but not limited to definitions described in the technical specification, are part of a greater international controlled terminology resource managed by an internationally recognized standards development organization (SDO). CDISC has been identified by ICH as a reputable SDO with the qualifications and capabilities to support the maintenance and facilitation of the governance process for ICH controlled terminology.

This Memorandum of Understanding (MOU) sets forth the roles and responsibilities of each party as they relate to the governance of the ICH terms and definitions developed in collaboration with CDISC. This MOU is intended to describe the goals, the high-level governance process, and how each party will collaborate. Specific projects (e.g., M11 controlled terminology) will be defined in detail as part of an annex to this MOU mutually agreed upon by CDISC and ICH.

Goals

As a collaboration between ICH and CDISC, the goals of the agreement are to:

1. Use a unified governance process and terminology services for the long-term support of ICH controlled terminologies.
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4. Ensure the terminologies are freely available to the public following public review.

ICH harmonisation for better health

The graphic provides a visual guide to how the content defined by the ICH M11 Teamplate speciciation is handled within USDM

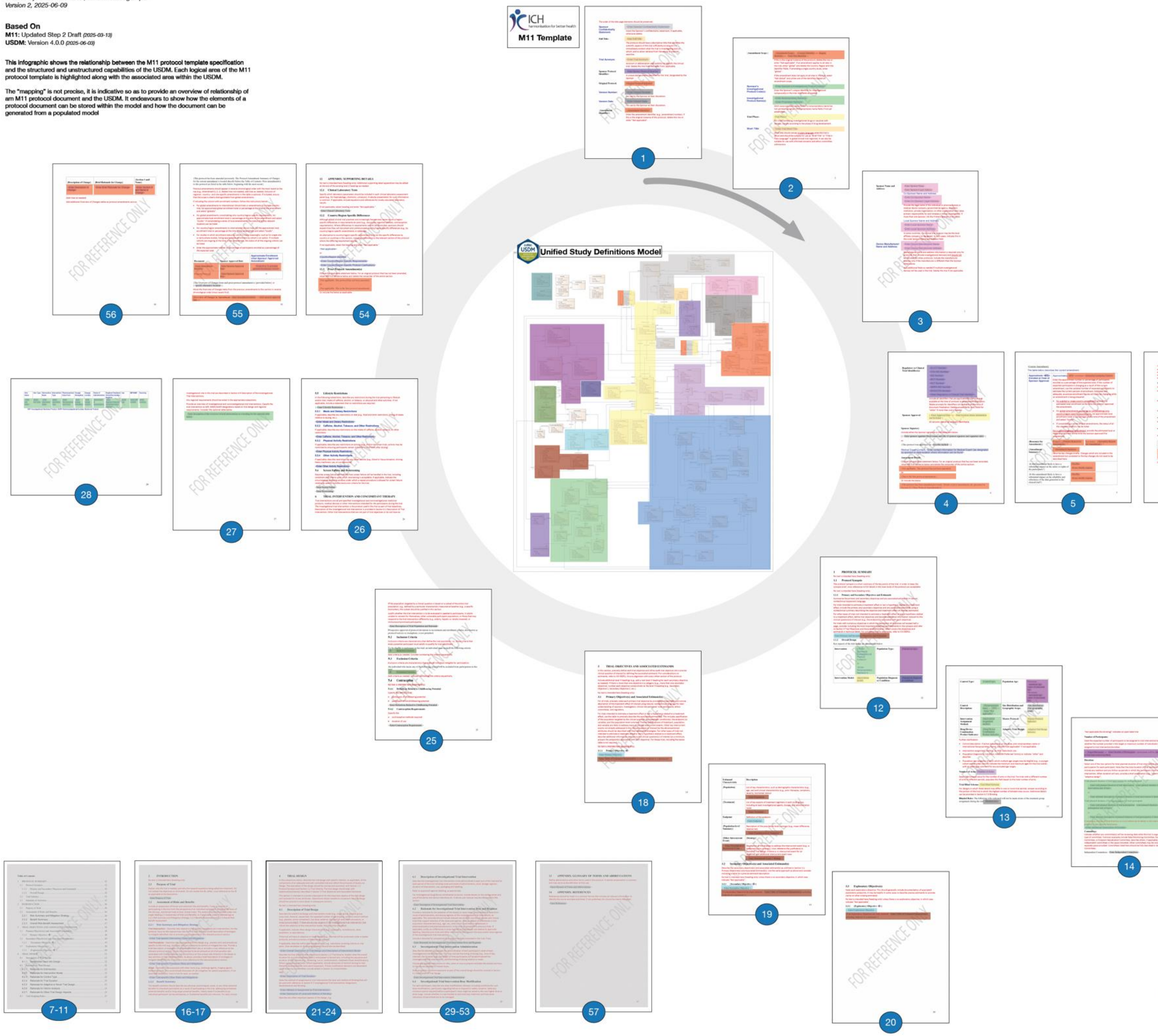
M11 and USDM Alignment

Created by: D Iberson-Hurst, data4knowledge ApS
Version 2, 2025-06-09

Based On
M11: Updated Step 2 Draft *(2025-03-13)*
USDM: Version 4.0.0 *(2025-06-03)*

This infographic shows the relationship between the M11 protocol template specification and the structured and unstructured capabilities of the USDM. Each logical area of the M11 protocol template is highlighted along with the associated area within the USDM.

The "mapping" is not precise, it is indicative so as to provide an overview of relationship of am M11 protocol document and the USDM. It endeavours to show how the elements of a protocol document can be stored within the model and how the document can be generated from a populated model



Key

Indicates the page number within the ICH M11 Template Specification

N

- Organizations
- Identifiers
- Amendments
- Titles
- Objectives & Endpoints
- Inclusion & Exclusion
- Population & Cohorts
- Estimands
- Interventions
- Study & Study Design
- High Level Study Design
- Detailed Study Design
- Indications
- Documents and Narrative Content
- Roles
- Controlled Terminology

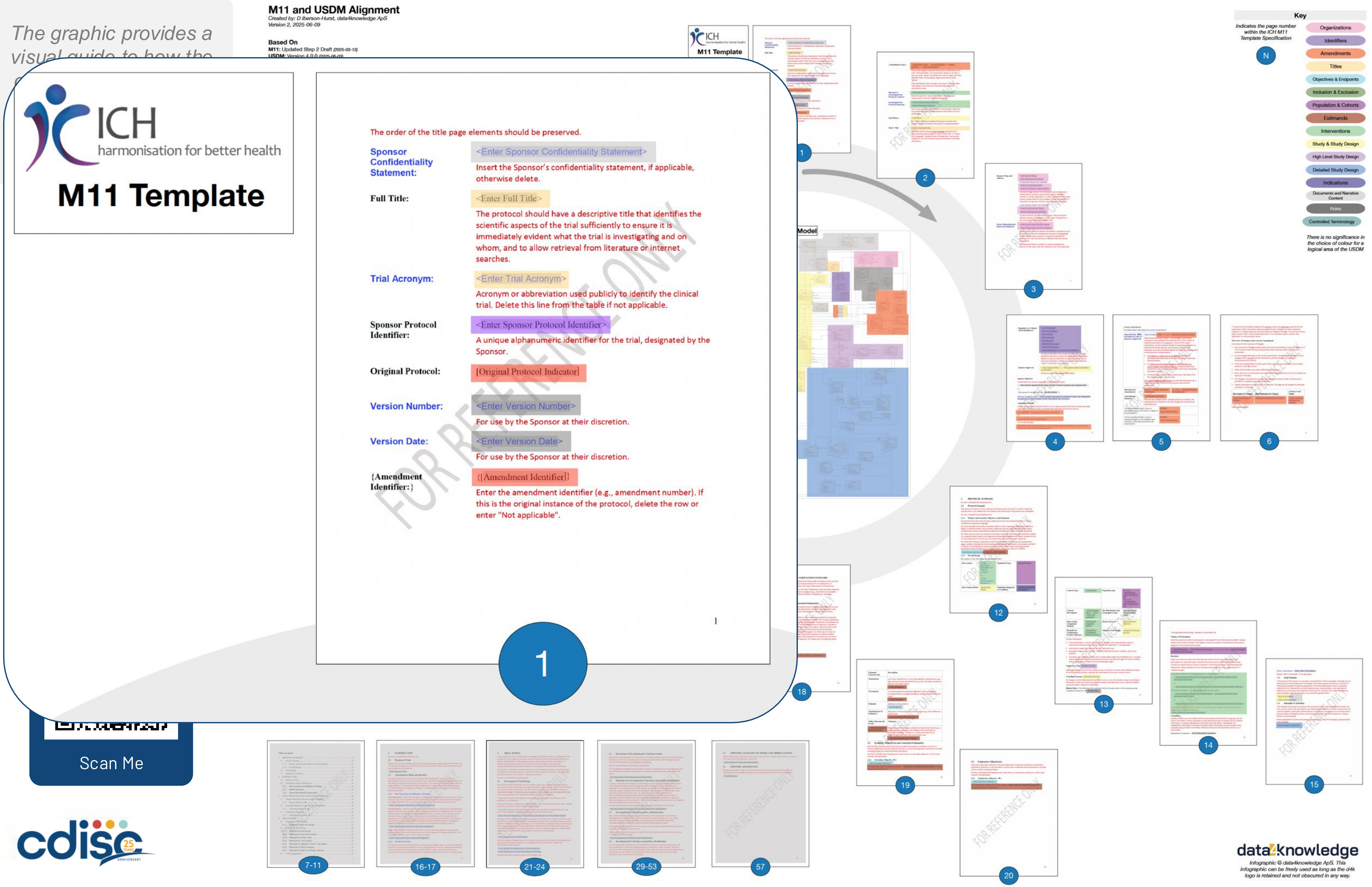
There is no significance in the choice of colour for a logical area of the USDM

USDM <-> M11

Scan Me

data4knowledge

The graphic provides a visual guide to how the



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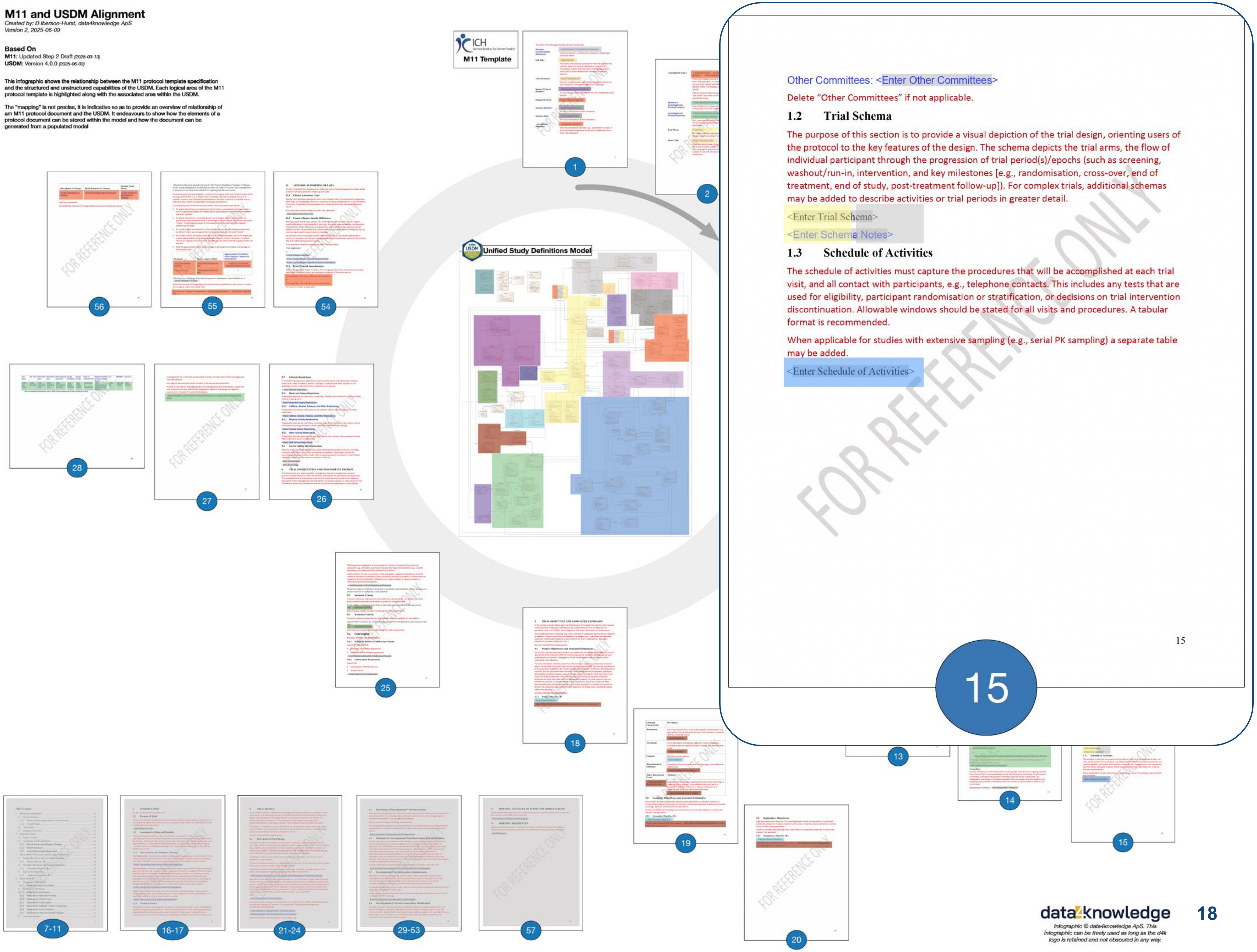


USDM <-> M11

Scan Me

data4knowledge

HL7 FHIR Connectathon

Dallas, May 2024

- Brian Alper, Computable Publishing
- Matt Elrod, ONC
- Hugh Glover, Vulcan ***TRACK LEAD***
- Dave Iberson-Hurst, CDISC
- Dan Newingham, ZS Associates
- Jimita Parekh, ICH M11/M2
- Khalid Shahin, Computable Publishing
- Rik Smithies, HL7 UK
- Kimberly Tableman, Espero
- Panagiotis Telonis, ICH M11/M2
- Stacy Tegan, Vulcan, TransCelerate
- Rob Ferendo, TransCelerate

ICH M11/M2

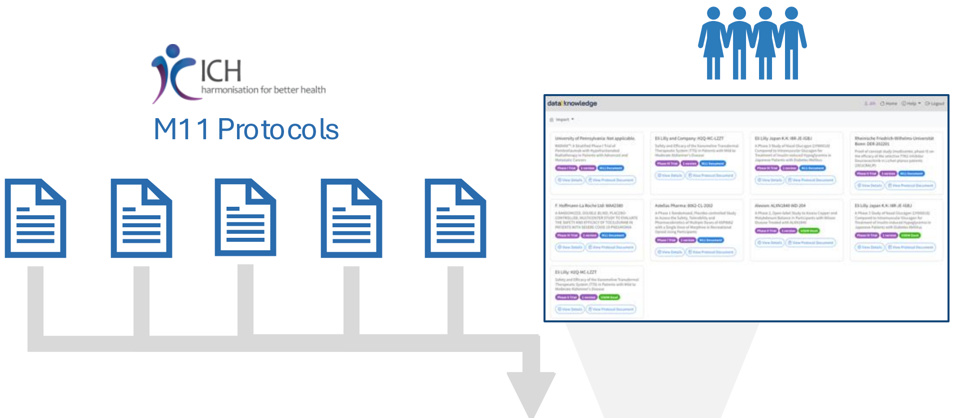
HL7 FHIR Connectathon

Atlanta, September 2024

- Brian Alper, Computable Publishing
- Hugh Glover, Vulcan *TRACK LEAD*
- Dave Iberson-Hurst, CDISC
- Khalid Shahin, Computable Publishing
- Rik Smithies, HL7 UK
- Stacy Tegan, Vulcan, TransCelerate
- Rob Ferendo, TransCelerate
- Mary Lynn Mercado, Novartis
- Enam Rahman, TransCelerate
- Mihoko Okada, Idial

ICH harmonisation for better health
M2 / M11

ICH harmonisation for better health
M11 Protocols



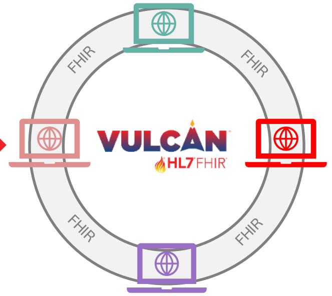
"Sponsor" 1

- M11 '.docx' → USDM → FHIR

data4knowledge

M11 Protocol | Dashboards
Unified Study Definitions Model
FHIR M11 Message / Profile

FHIR M11 Message / Profile



FHIR M11 Message / Profile

FHIR Viewer
NProgram Ltd

"Regulator" 2

- FHIR → USDM → M11 Presentation

Structure parts of the M11 Template:

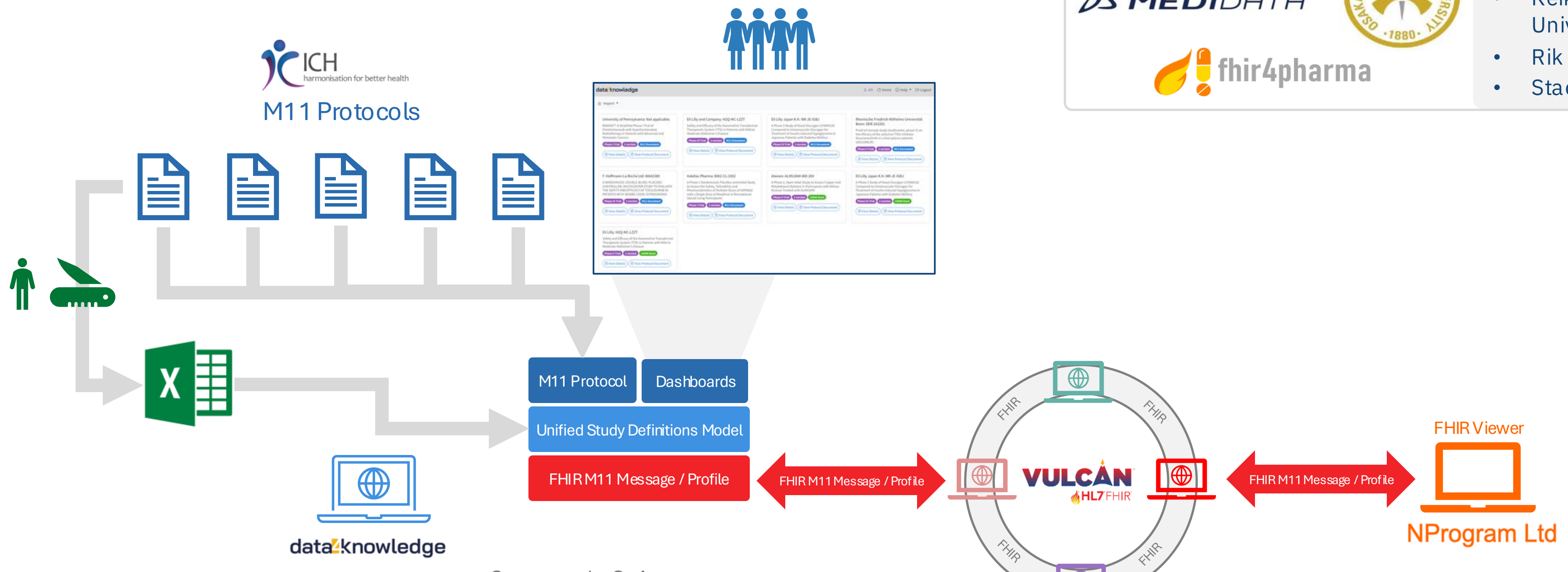
- *Title Page*
- *Inclusion / Exclusion*
- *Estimands, Objectives and Endpoints*
- *Amendments*

M11 Protocol	Dashboards	**Human / Business Layer** M11 Protocol Template and Technical Specification
Unified Study Definitions Model		**System / Database Layer** Unified Study Definitions Model
FHIR M11 Message		**Communication Layer** FHIR M11 message / profile (JSON or XML)

HL7 FHIR Connectathon

Virtual, January 2025

- Hugh Glover, Vulcan TRACK LEAD
- Rob Ferendo, TransCelerate
- Patrick Genyn, fhir4pharma
- Dave Iberson-Hurst, CDISC
- Geoff Low, Medidata
- Keiko Ota, Osaka Metropolitan University Hospital
- Rik Smithies, HL7 UK
- Stacy Tegan, TransCelerate, Vulcan



Structure the SoA

- *Visits*
- *Activities*
- *Timing*

"Sponsor" 1

- M11 '.docx' → USDM → FHIR M11
- USDM '.xlsx' → USDM → FHIR SoA

"Regulator" 2

- FHIR → M11 Presentation

"Operational Environment" 3

- FHIR → SoA Presentation

M11 Protocol	Dashboards	**Human / Business Layer** M11 Protocol Template and Technical Specification
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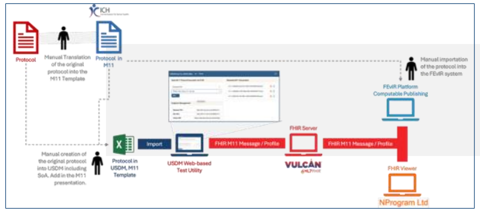
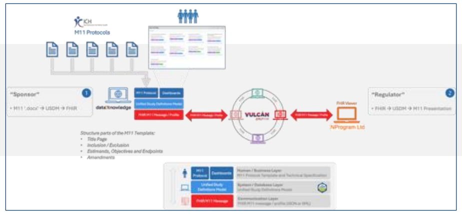
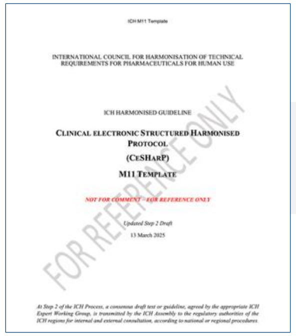
HL7 FHIR Connectathon

Madrid, May 2025

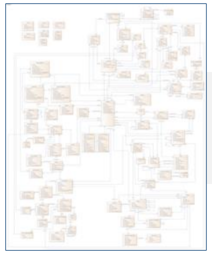
Dallas & Atlanta Connectathons



ICH M11 Template & Technical Specifications

ICH M11 and CDISC USDM share the same CT

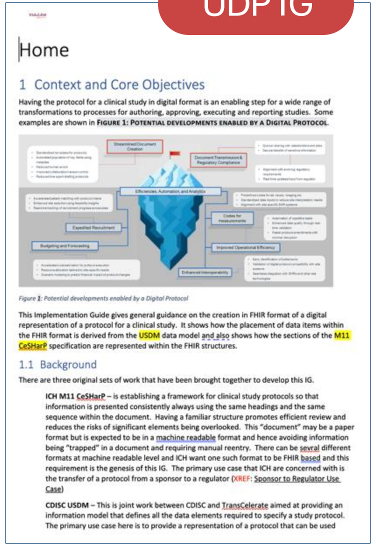
Controlled Terms



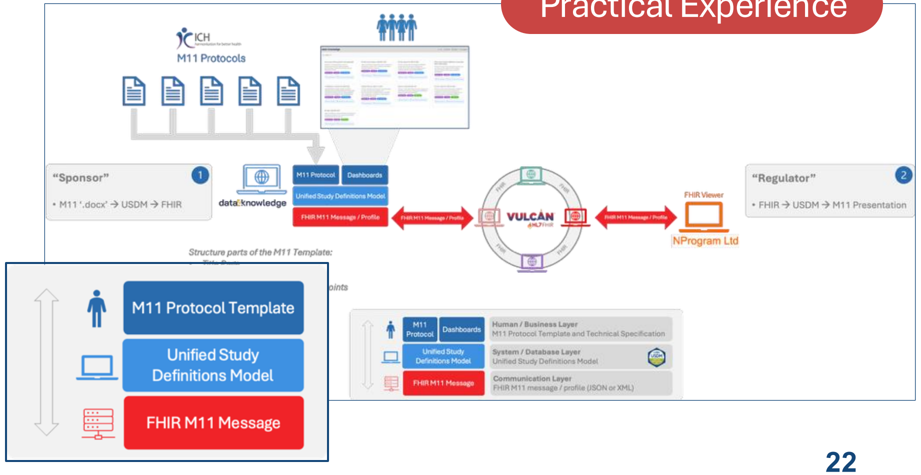
CDISC USDM

Logical Model

UDP IG



Practical Experience



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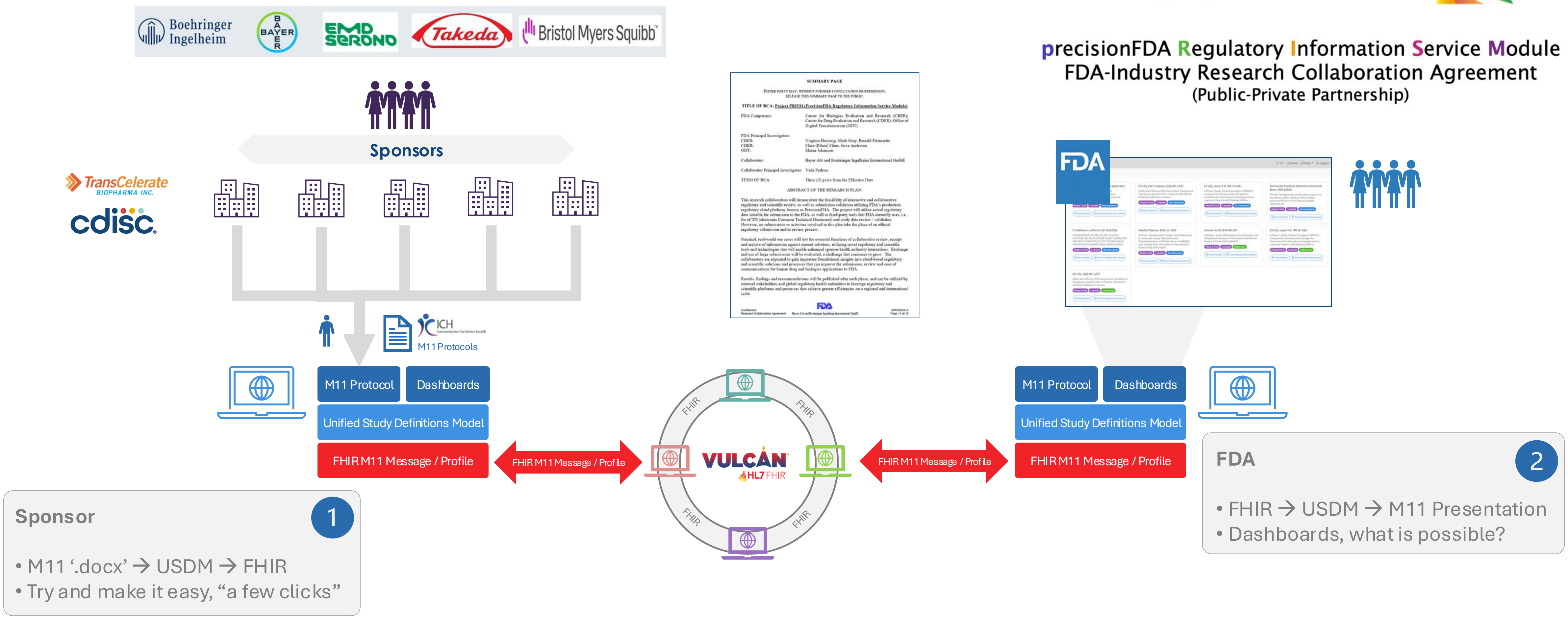
PROJECT PRISM

A Regulatory Cloud Collaborative Initiative

precisionFDA Regulatory Information Service Module
FDA-Industry Research Collaboration Agreement
(Public-Private Partnership)



M11 Protocol	Dashboards	**Human / Business Layer** M11 Protocol Template and Technical Specification
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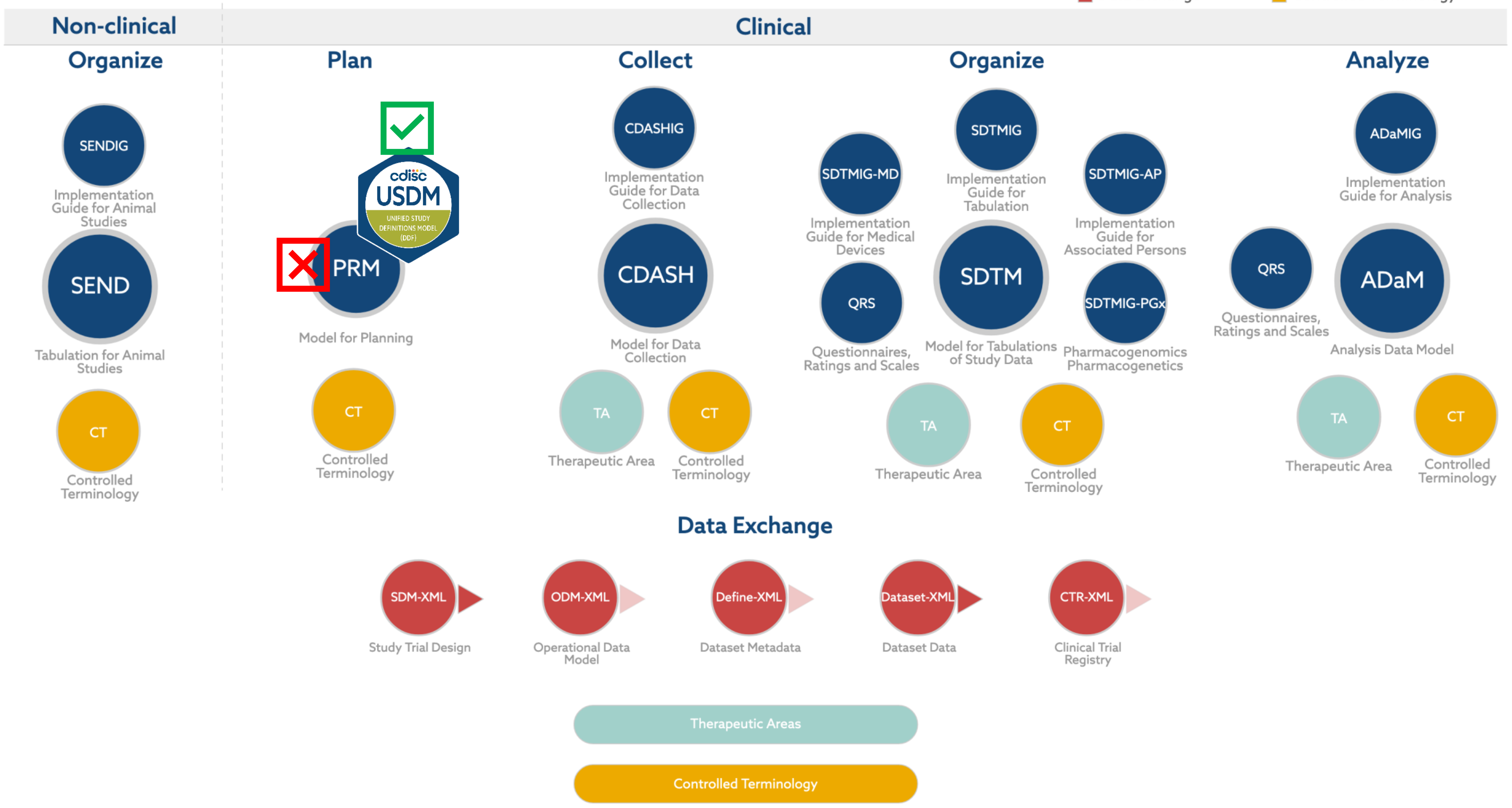
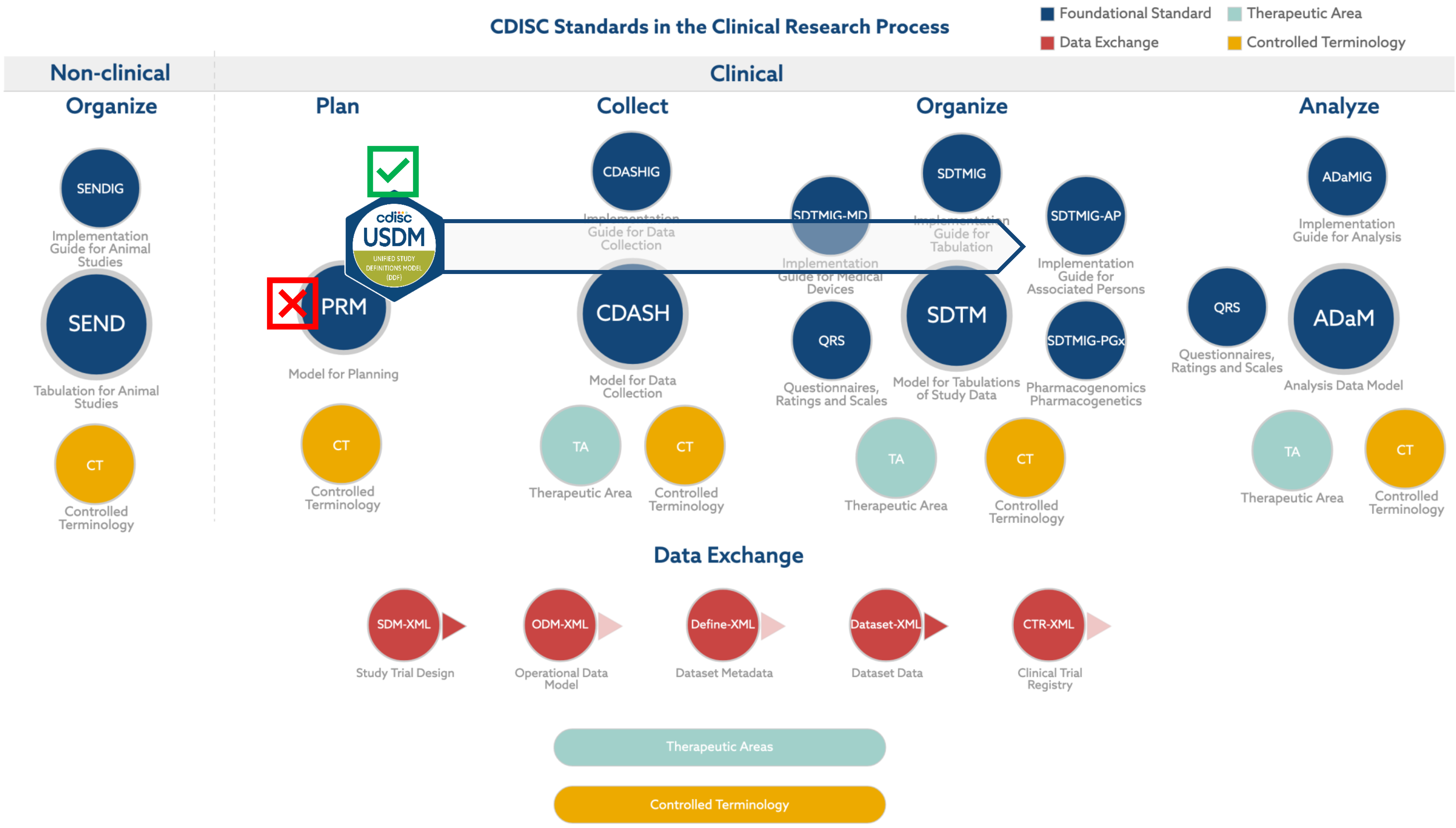
How do the Pieces Fit?

CDISC Standards in the Clinical Research Process

Foundational Standard
Therapeutic Area
Data Exchange
Controlled Terminology



CDISC Standards in the Clinical Research Process



USDM in Action
Use Cases Supporting the DDF Vision

Use Cases

Retrospective

Protocol 'Store'

Past Protocols
The storage of past sponsor protocols in to support a variety of use cases such as:

- Therapeutic area specific content
- Standard inclusion and exclusion criteria.
- Libraries of objectives, endpoints and estimands.
- Asses the past to prevent protocol amendments.
- General search across the library

Prospective

Study Design

Protocol Authoring
Support the authoring of new protocols including complex studies such as Master, Umbrella etc.

This would include analyzing previous protocol information to inform design choices (see Protocol Store).

Writing associated SAP, Informed Consent, Monitoring Plan, Drug Plan etc via content reuse.

Study Build
Build of the study based on the protocol l including all detailed procedures and assessments to allow for immediate deployment.

Feasibility & Cost
Support study feasibility and complexity evaluations.

Determine the cost of the study including such items as vendor and CRO costs and support the laboratory RFP process

Regulatory Authority
Approval for the execution of the study including IND, CTA etc and discussions with regulatory authorities

Stakeholder Views
Tailored views for stakeholders such as IRB and ethic committees.

Study Start-up

Deployment
Deployment of the study across the enterprise using the USDM to support electronic configuration rather than manual data entry including Data collection systems (incl. EDC), CTMS, IRT, Data transmission specifications and more

TMF
Linking the protocol to the specification of milestones and essential documents

Data Collection Strategy
Ensure all data that is needed for analysis is collected. Remove unnecessary procedures & data collection to reduce trial cost and patient burden.

Trial Registries
Registration of the study with registries such as CT.gov, CTIS etc.

Training
Provision of training materials to those involved in the study including sites

Site Support
Provision of tailored information to sites regarding study execution.

Study Execution

Amendments
Support the amendment process with the ability to indicate the precise changes to the user community.

Ensure fewer amendments (see Protocol Store).

Subject Information
Provision of improved information to subjects regarding study execution, improve expectation management.

Drug & Study Materials
Assist in the planning and provision of study drug, materials and supplies

Subject Recruitment
Support the identification of subjects to help improve subject recruitment to the study. Help support EHR participation identification.

Analysis & Reporting

Automation
Provide a solid foundation for the automation of analysis and reporting outputs such as TLFs and CSRs.

SDTM
Creation of 'T' domains. Auto generation of data domains

ADaM
Use the protocol statistical metadata to assist in the derivation of analysis datasets

Statistical Analysis Plan
Use the protocol as the initial source of the SAP to ensure consistency with objectives, endpoints, estimands

Regulatory Submission

Submission
Preparation of the submission by the sponsor

Regulatory Review
Review of the submission by the regulatory authority and discussions / questions with the sponsor

Statistical and Safety Review
Support the comparison of data by arm or study design.

Allow for the verification of planned (protocol) data versus actual (study data) captured data

Across The Lifecycle

Dashboards
The display of protocol related information sourced from one or more protocols in ways which have not been achievable before

Comparison
Compare and verify differences across versions and similar protocols

Unified Study Definitions Model (USDM)

- Controlled Terms
- Unstructured Content
- Populations, Inclusion & Exclusion, Interventions & Indications, Estimands, Objectives & Endpoints
- Study, Identifiers, Amendments
- Study Designs, Arms, Epochs
- Procedures, Biomedical Concepts
- Detailed Study Logic, Encounters

Study Details
The overall study, its various versions, identifiers and associated governance. Also includes the amendments made to the study

Study High Level Design
The single or set of study designs making up the study detailing the epochs, arms etc.

Study Science
The detailed description of the study science: the populations and the associated inclusion and exclusion criteria, the indications being studied, the interventions being used and the objectives, endpoints and the associated estimands.

Detailed Study Logic
A precise definition of the study logic including support for the Schedule of Activities.

Unstructured Content
The ability to support one or more document presentations of the USDM content including the ICH M11 protocol template, sponsor templates and other documents.

Procedures and Biomedical Concepts
The detail around the procedures and observations to be performed as part of the detailed study designs.

Controlled Terms
The controlled terminology needed to define the semantics within the model. Managed in the same manner as all CDISC CT and aligned with the M11 template standard.

Use Cases

Scan Me

NOTE: The use cases presented are illustrative and the list is not intended to be exhaustive.

Version 5, 24th October 2024. Prepared by D Iberson-Hurst for the TransCelerate 'DDF in Action' day.
With thanks to Rob Ferendo (TransCelerate), Bill Illis (Novartos), Jasmine Kestemont (Argenx), Kirsten Langendorf (d4k), Mary Lynn Mercado (Novartis), Lissa Morgan (Amgen), Johannes Ullander (d4k) and Peter Van Reusel (CDISC)

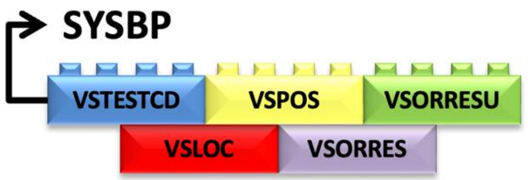
BCs Downstream?

Sometime ago …

DIA Annual, 2009

Looking at the meta data …

USUBJID	VSSEQ	VSTESTCD	VSTEST	VSPOS	VSLOC	VSORRES	VSORRESU
ABC-001-001	12	SYSBP	Systolic Blood Pressure	SITTING	LEFT ARM	95	mmHg
ABC-001-001	13	DIABP	Diastolic Blood Pressure	SITTING	LEFT ARM	44	mmHg



In the same way as with lego we can plug the bricks together, so systolic blood pressure can be described by a collection of meta data bricks

Metadata …

FDA's STUDY DATA TECHNICAL CONFORMANCE GUIDE

Contains Nonbinding Recommendations

Appendix: Data Standards and Interoperable Data Exchange

This appendix provides some of the guiding principles for the Agency's long-term study data standards management strategies. An important goal of standardizing study data submissions is to achieve an acceptable degree of *semantic interoperability* (discussed below). This appendix describes different types of interoperability and how data standards can support interoperable data exchange now and in the future.

At the most fundamental level, study data can be considered a collection of data elements and their relationships. A data element is the smallest (or *atomic*) piece of information that is useful for analysis (e.g., a systolic blood pressure measurement, a lab test result, a response to a question on a questionnaire).

A data value is by itself meaningless without additional information about the data (so called *metadata*). Metadata is often described as *data about data*. Metadata is structured information that describes, explains, or otherwise makes it easier to retrieve, use, or manage data.[48] For example, the number *44* itself is meaningless without an association with Hematocrit and the unit of measurement (e.g. "%"). Hematocrit in this example is metadata that further describes the data.

Just as it is important to standardize the representation of data (e.g., M and F for male and female, respectively), it is equally important to standardize the metadata. The

of metadata. Data about a library book such as author, type of book, and the Library of Congress number, are metadata and were once maintained on index cards. SAS labels and formats are a rudimentary form of metadata, although they have not historically been referred to as metadata.

31

December, 2014

Contains Nonbinding Recommendations

U.S. FOOD & DRUG ADMINISTRATION

Appendix A: Data Standards and Interoperable Data Exchange

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Just as it is important to standardize the representation of data (e.g., M and F for male and female, respectively), it is equally important to standardize the metadata. The expressions Hematocrit = 44; Hct = 44, or Hct Lab Test = 44 all convey the same information to a human, but an information system or analysis program will fail to recognize that they are

frequently used to refer to XML (extensible markup language) tags, there is nothing new about the concept of metadata. Data about a library book such as author, type of book, and the Library of Congress number, are metadata and were once maintained on index cards. SAS labels and formats are a rudimentary form of metadata, although they have not historically been referred to as metadata.

U.S. Food & Drug Administration
10903 New Hampshire Avenue
Silver Spring, MD 20903
www.fda.gov

Page 68 of 88

March, 2025

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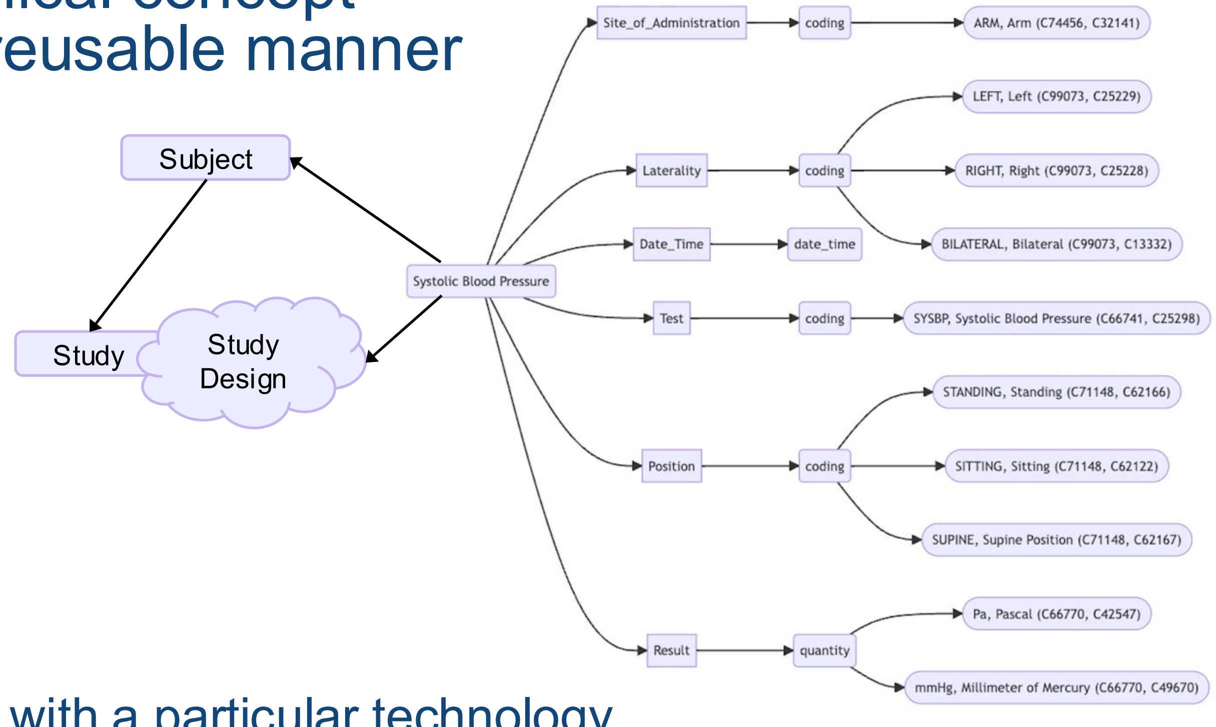
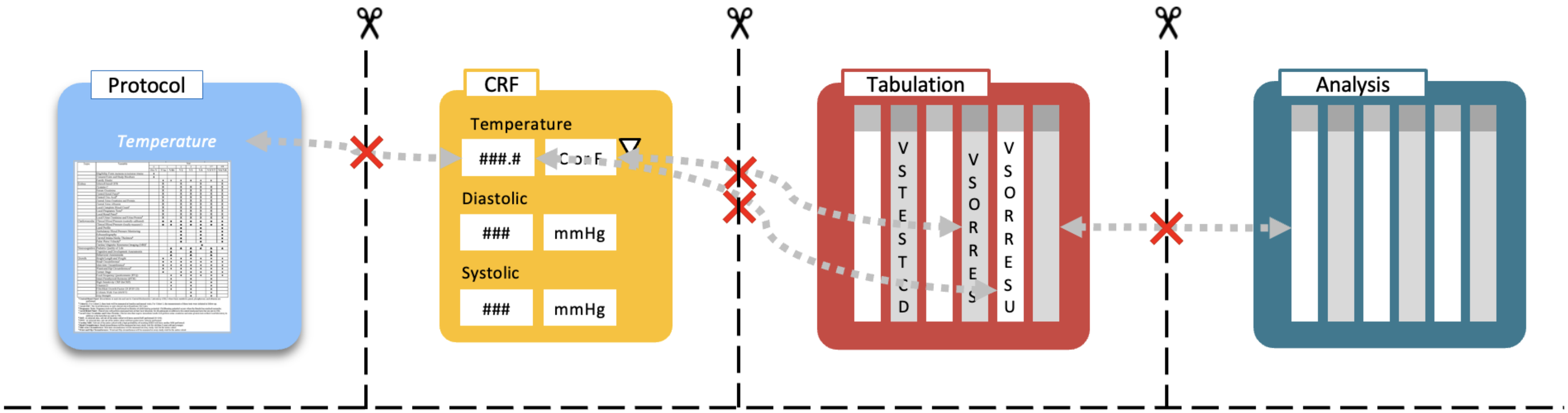
A Biomedical Concept is …

- A small data model that defines a clinical concept (observation) in a standardized and reusable manner
- Atomic:
  - If it is split, it loses meaning
- Identifiable:
  - Has a unique identifier
  - Find it, reference it, deploy it
- Complete:
  - Everything is defined
- Data Specification
  - Specification of the data, not how it is used with a particular technology
- Context:
  - A BC needs context, i.e. the rest of the DDF model, a study, the encounters, activities, timing …



Why BCs … Silo Killers

PHUSE, 2019



Complexity

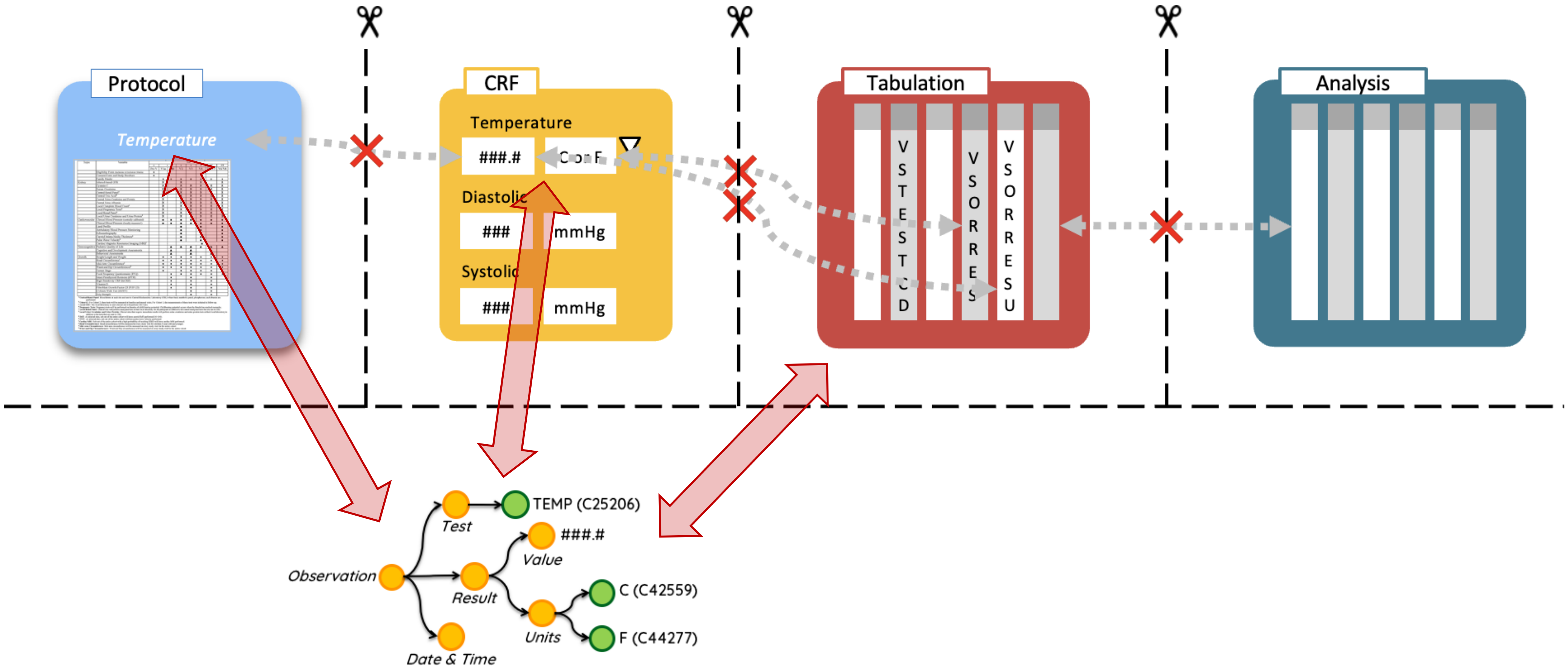
When standards were originally developed the space is so complex we split it up into manageable pieces … But we created silos

CDISC standards are what we want, as humans, to see.

Why BCs … Silo Killers

Machine / Data Layer

We need to remove the silos inside of the machine, give the machine more information, more context. We need to think about it from a machine / data perspective.

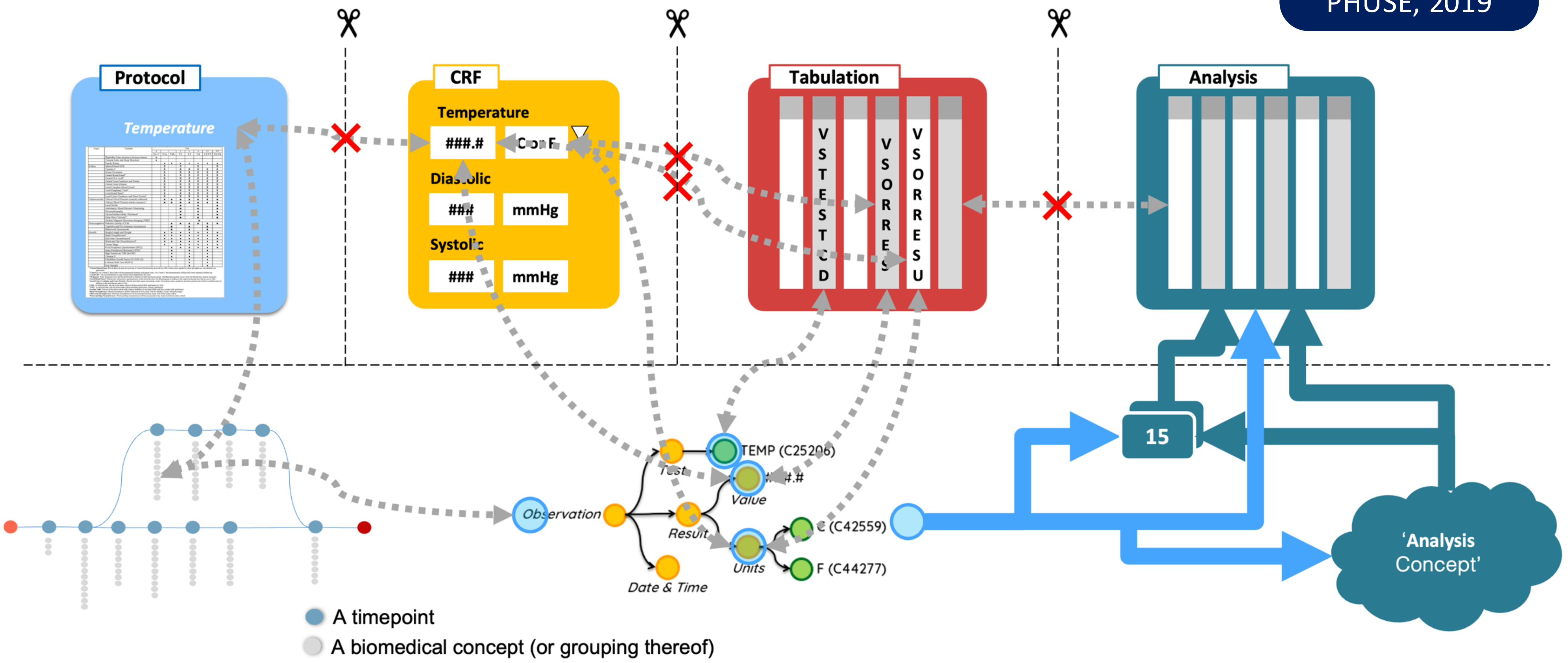


Machine / Data Layer

Machine / Data Layer

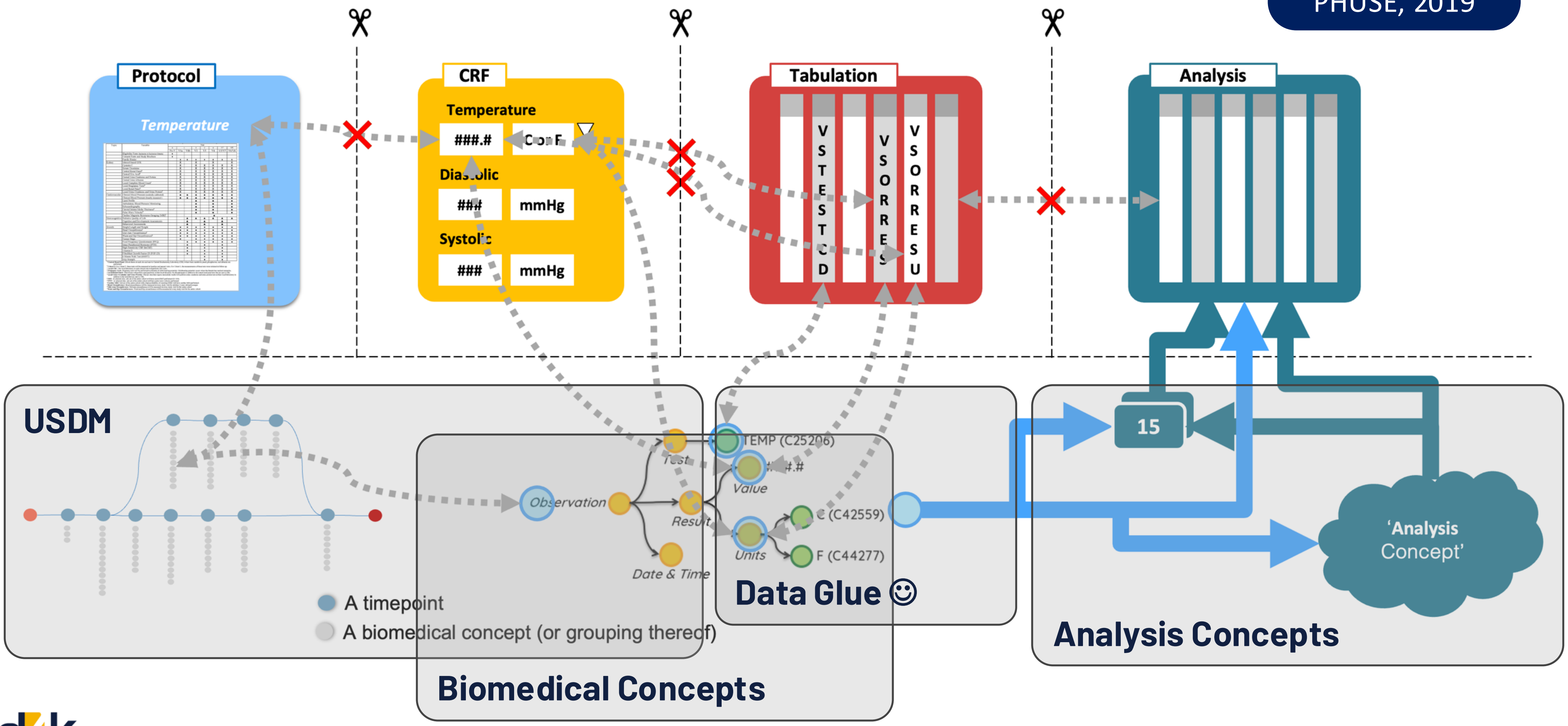
We need to remove the silos inside of the machine, give the machine more information, more context, put data at the centre.

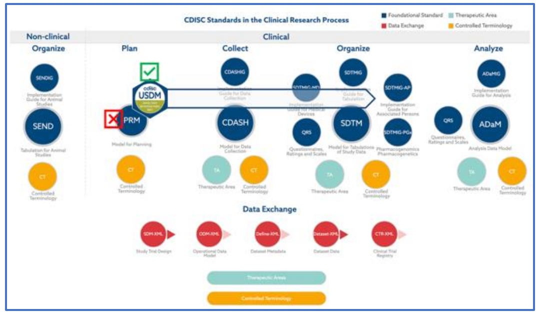
PHUSE, 2019



Machine / Data / Model Layer

PHUSE, 2019





Typical Protocol

Typical Protocol

SoA provides the activities and then references "labels" in footnotes and/or text detailing the observations required and various conditions.

Those "labels" are the names of BCs

USDM can provides that raft of information in a structured manner

Appendix 2
Schedule of Activities: Days 3–28

	Days 3–28 [a]																										Study Completion/ Discontinuation
Study Day	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
Chest X-ray/CT scan					x							x							x							x	x
Vital signs [b]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
PaO_2/FiO_2 [c]	← Optional →																										Optional
SpO_2 [b]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Ordinal scoring [d]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Adverse events [e]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Concomitant medications [f]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Hematology [g]	x				x			x				x							x							x	x
Chemistry [h]	x				x			x				x							x							x	x
Central Labs																											
Serum PD (CRP, IL-6, sIL-6R)	x				x							x							x							x	x
Serum PK					x							x							x							x	x
Serum sample for exploratory biomarkers	x				x							x							x							x	x
SARS-CoV-2 viral load [i]	x	x	x	x	x			x				x							x							x	x
Serum SARS-CoV-2 antibody titer																										x	x
Cryopreserved PBMCs [j]	x				x							x							x							x	x

[b] All vital sign measurements (i.e., respiratory rate, pulse rate, systolic and diastolic blood pressures, and body temperature), oxygen saturation and NEWS2-specific assessments (i.e., consciousness and presence or absence of oxygen support) should be recorded together twice daily with approximately 12 hours in between while the patient remains hospitalized. If measured more than once during a 12-hour period, the worst values (highest temperature, respiratory rate, and heart rate, lowest blood pressure, oxygen saturation, and consciousness level) during that period should be recorded on the eCRF. Following hospital discharge these parameters should be recorded once at each return visit to the clinic. Vital signs and oxygen saturation will not be recorded if follow-up visits are conducted by telephone.

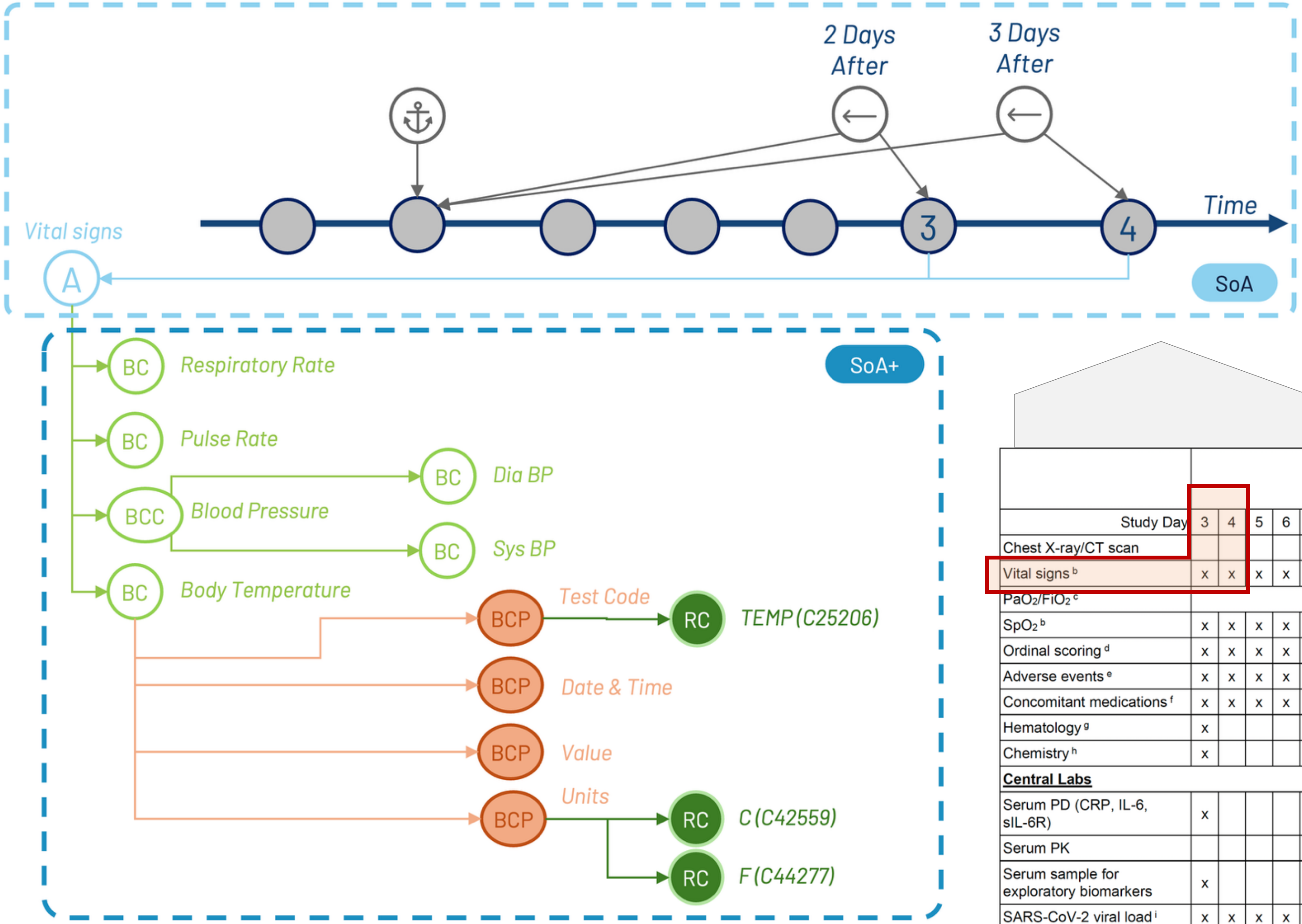
4.5.4 Vital Signs and Oxygen Saturation

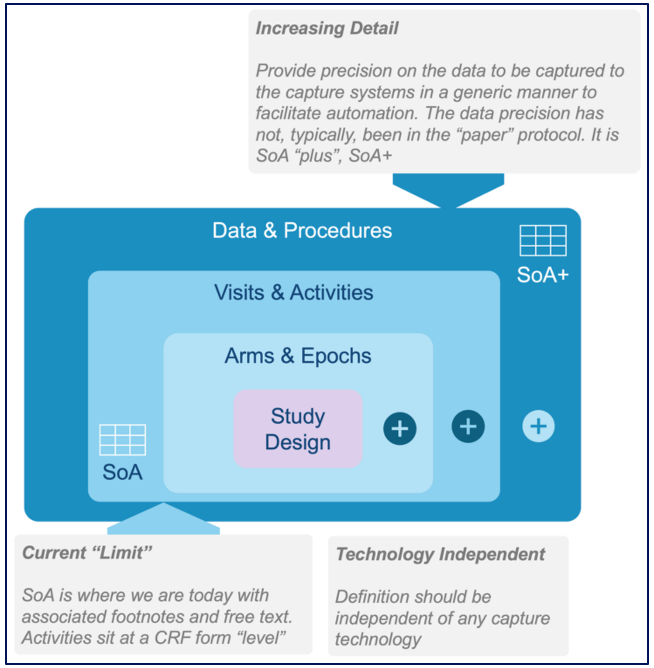
Vital signs will include measurements of respiratory rate, pulse rate, and systolic and diastolic blood pressure, and body temperature. Peripheral oxygen saturation should also be measured at the same time as the vitals. For patients requiring supplemental oxygen, the oxygen flow rate (L/min) and/or fraction of inspired oxygen (FiO2) should be recorded.

USDM & SoA+

SoA and SoA+

SoA is the information we typically see in the SoA table in a paper protocol. SoA+ is the extra detail defining the observations. SoA+ is the linking in of the BCs to provide the information found in the footnotes etc.





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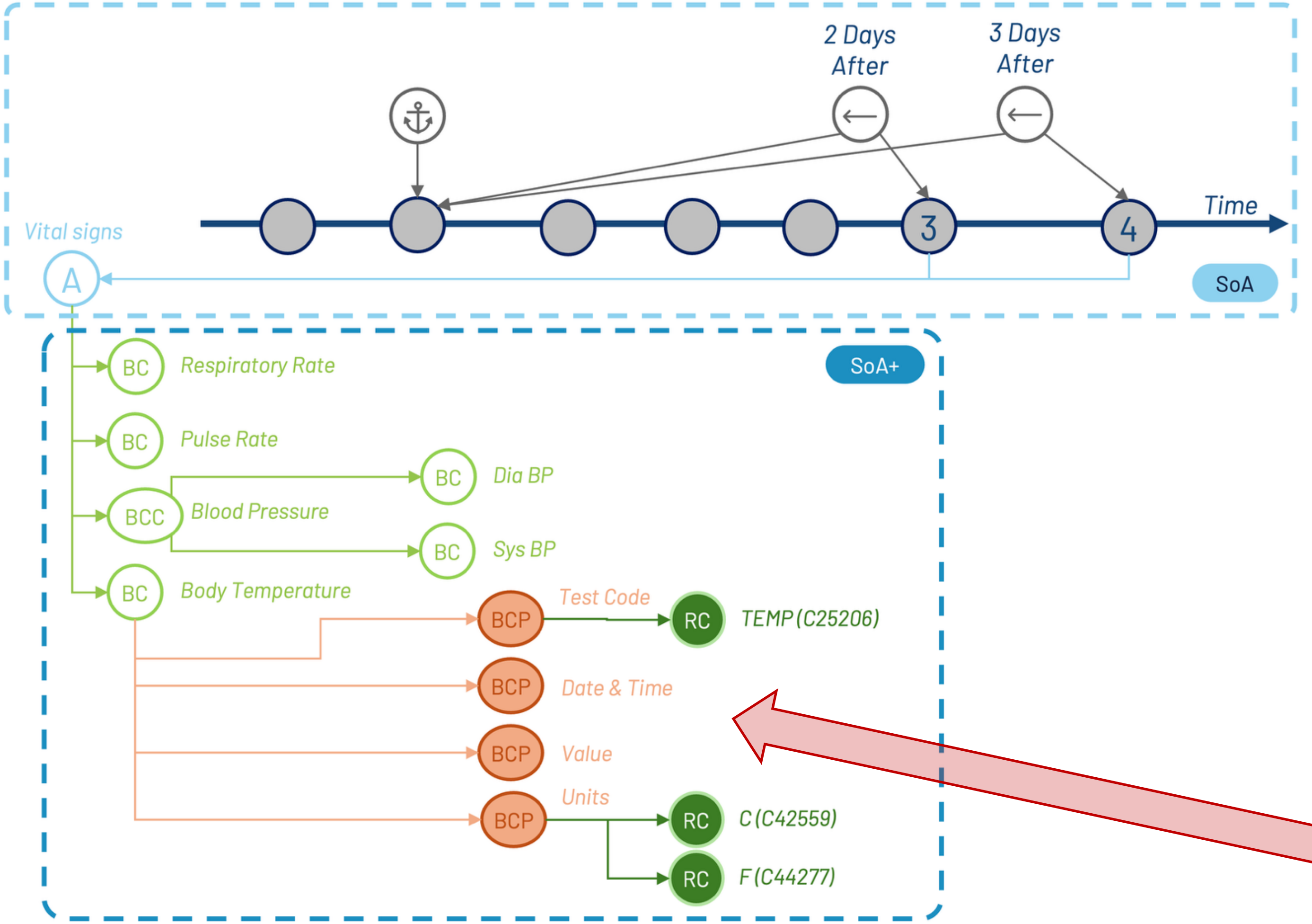
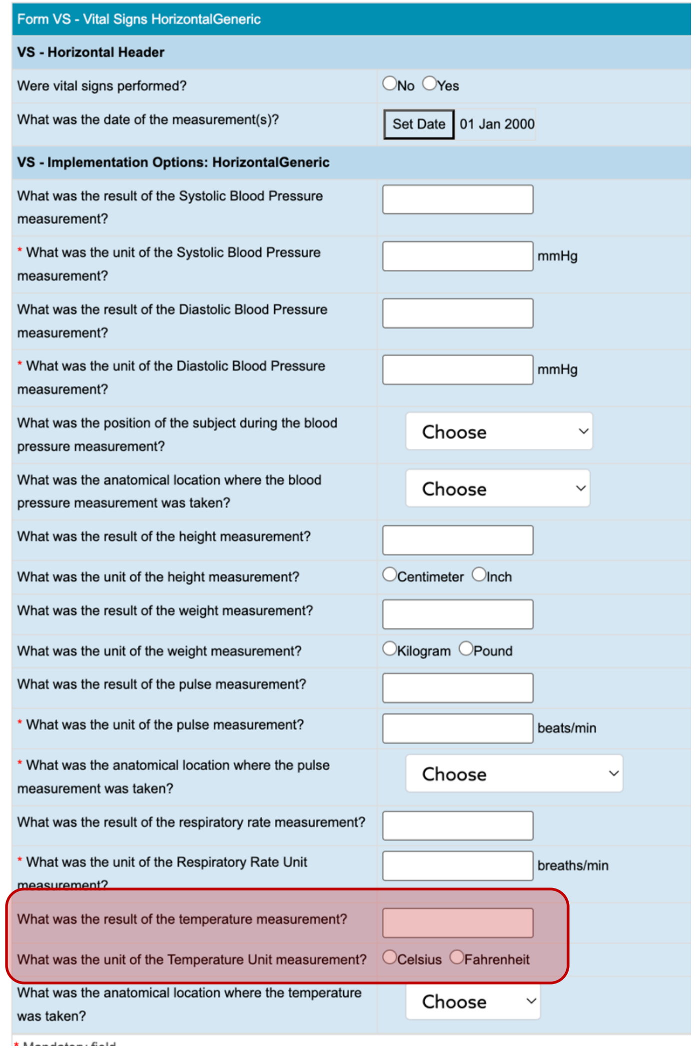
Chest X-ray/CT scan					x							x							x							x	x
Vital signs [b]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
PaO_2/FiO_2 [c]	← Optional →																										Optional
SpO_2 [b]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Ordinal scoring [d]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Adverse events [e]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Concomitant medications [f]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Hematology [g]	x				x			x				x							x							x	x
Chemistry [h]	x				x			x				x							x							x	x
Central Labs																											
Serum PD (CRP, IL-6, sIL-6R)	x				x							x							x							x	x
Serum PK					x							x							x							x	x
Serum sample for exploratory biomarkers	x				x							x							x							x	x
SARS-CoV-2 viral load [i]	x	x	x	x	x			x				x							x							x	x
Serum SARS-CoV-2 antibody titer																										x	x
Cryopreserved PBMCs [j]	x				x							x							x							x	x

1. Link to Existing CRF Library

“Traditional” Approach (Walk)

Take the BCs in the design and “map” to an existing forms library and work in a traditional manner for the remainder of the left-cycle as per today (e.g. SDTM).





Form VS - Vital Signs HorizontalGeneric

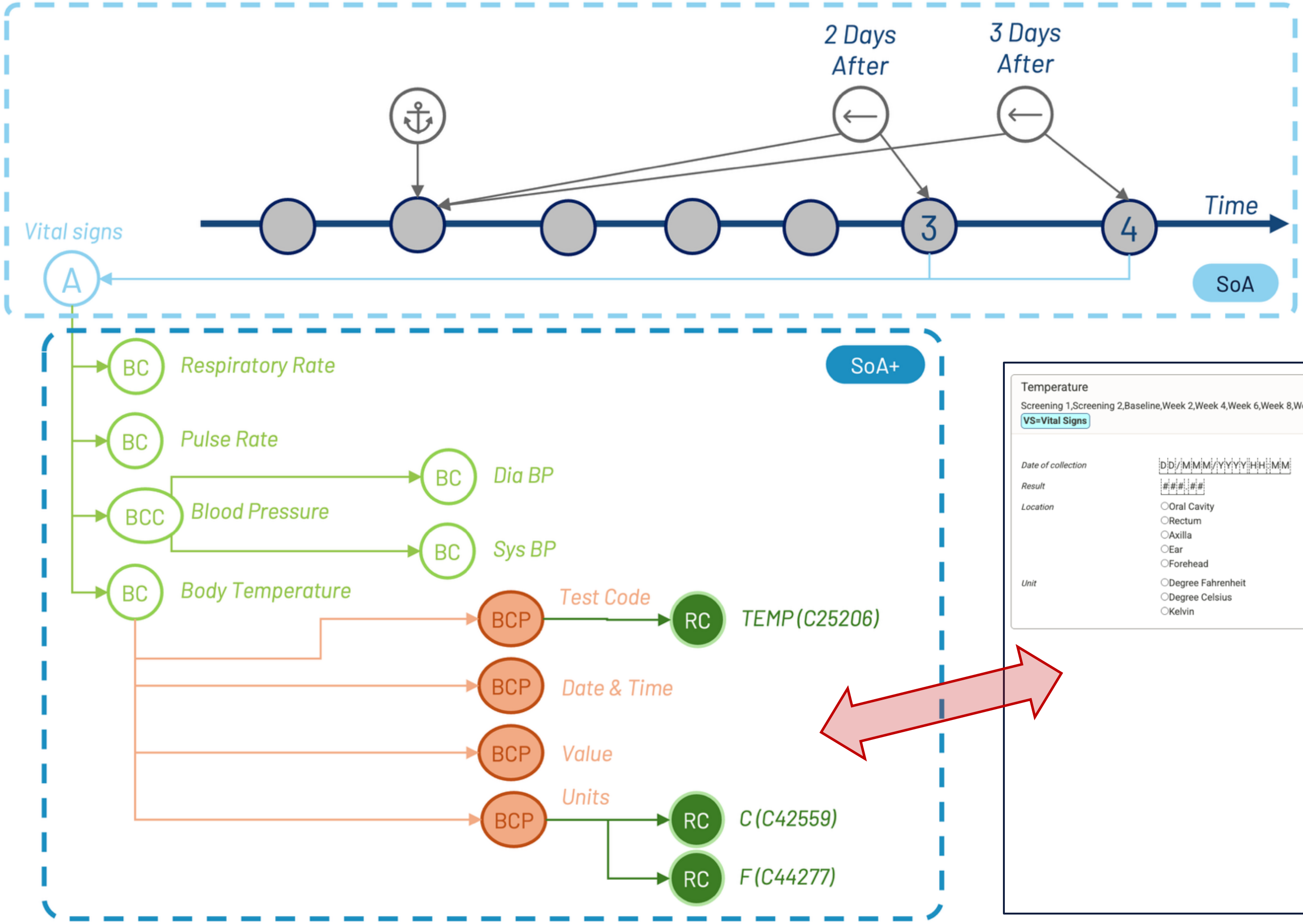
VS - Horizontal Header	
Were vital signs performed?	○No ○Yes
What was the date of the measurement(s)?	Set Date 01 Jan 2000
VS - Implementation Options: HorizontalGeneric	
What was the result of the Systolic Blood Pressure measurement?	
* What was the unit of the Systolic Blood Pressure measurement?	mmHg
What was the result of the Diastolic Blood Pressure measurement?	
* What was the unit of the Diastolic Blood Pressure measurement?	mmHg
What was the position of the subject during the blood pressure measurement?	Choose
What was the anatomical location where the blood pressure measurement was taken?	Choose
What was the result of the height measurement?	
What was the unit of the height measurement?	○Centimeter ○Inch
What was the result of the weight measurement?	
What was the unit of the weight measurement?	○Kilogram ○Pound
What was the result of the pulse measurement?	
* What was the unit of the pulse measurement?	beats/min
* What was the anatomical location where the pulse measurement was taken?	Choose
What was the result of the respiratory rate measurement?	
* What was the unit of the Respiratory Rate Unit measurement?	breaths/min
What was the result of the temperature measurement?	
What was the unit of the Temperature Unit measurement?	○Celsius ○Fahrenheit
What was the anatomical location where the temperature was taken?	Choose

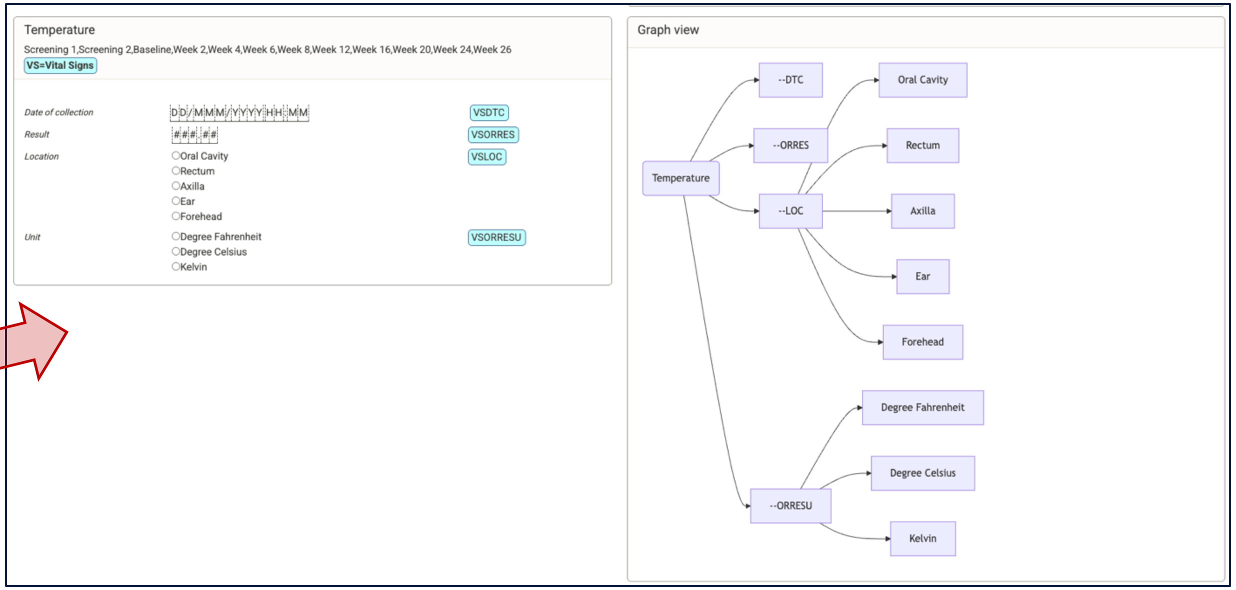
d4k

2. Generate Forms

More Adventurous Approach (Jog)

Take the BCs in the design and dynamically generate forms etc. Work in a traditional manner for the remainder of the left-cycle as per today (e.g. SDTM).

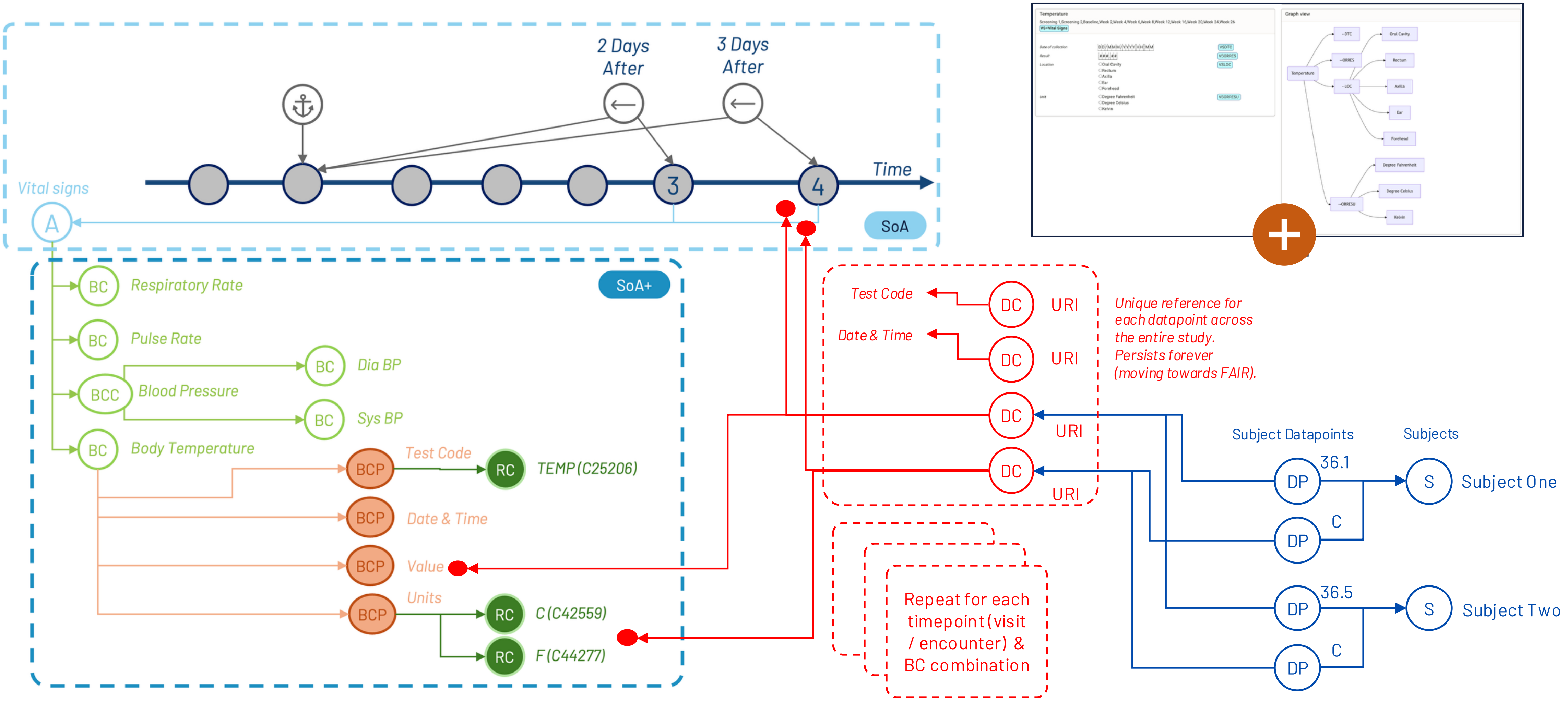




3. Data Context

Full Automation Approach (Run)

Use the USDM as the start of an operation model (extend the USDM) to bring design and data together. Generate the forms and link to the study definition. This also allows for auto creation of define.xml, aCRF, SDTM via linking in other models (not shown on this slide).



Benefits?

Benefits

Single Source of Truth

USDM's standardized framework eliminates data silos that have plagued clinical research by providing robust APIs for seamless data exchange across diverse platforms.

This ensures study data is consistently and reliable.

Reduce Cycle-Time

USDM can help reduce cycle times across the clinical trial lifecycle through the use of automation.

Given the large number of activities within each study, saving just a few hours on each activity quickly accumulates to substantial time savings across the entire clinical trial process.

This enables organizations to increase their clinical trial capacity.

Automation

USDM & BCs enable systems like EDC, CTMS, RTSM etc to be automatically configured from a single, standardized study definition.

This eliminates the need for manual transcription from PDF protocols into multiple systems, significantly reducing human error and accelerating study startup times through the "write once, read many times" principle.

Regulatory Compliance

A shared data model fosters better communication and alignment not only with among sponsors, CROs, and vendors but also with regulators.

The combination of USDM & ICH M11 will facilitate dialogue with regulators across the globe.

AI Enhancement Through Context

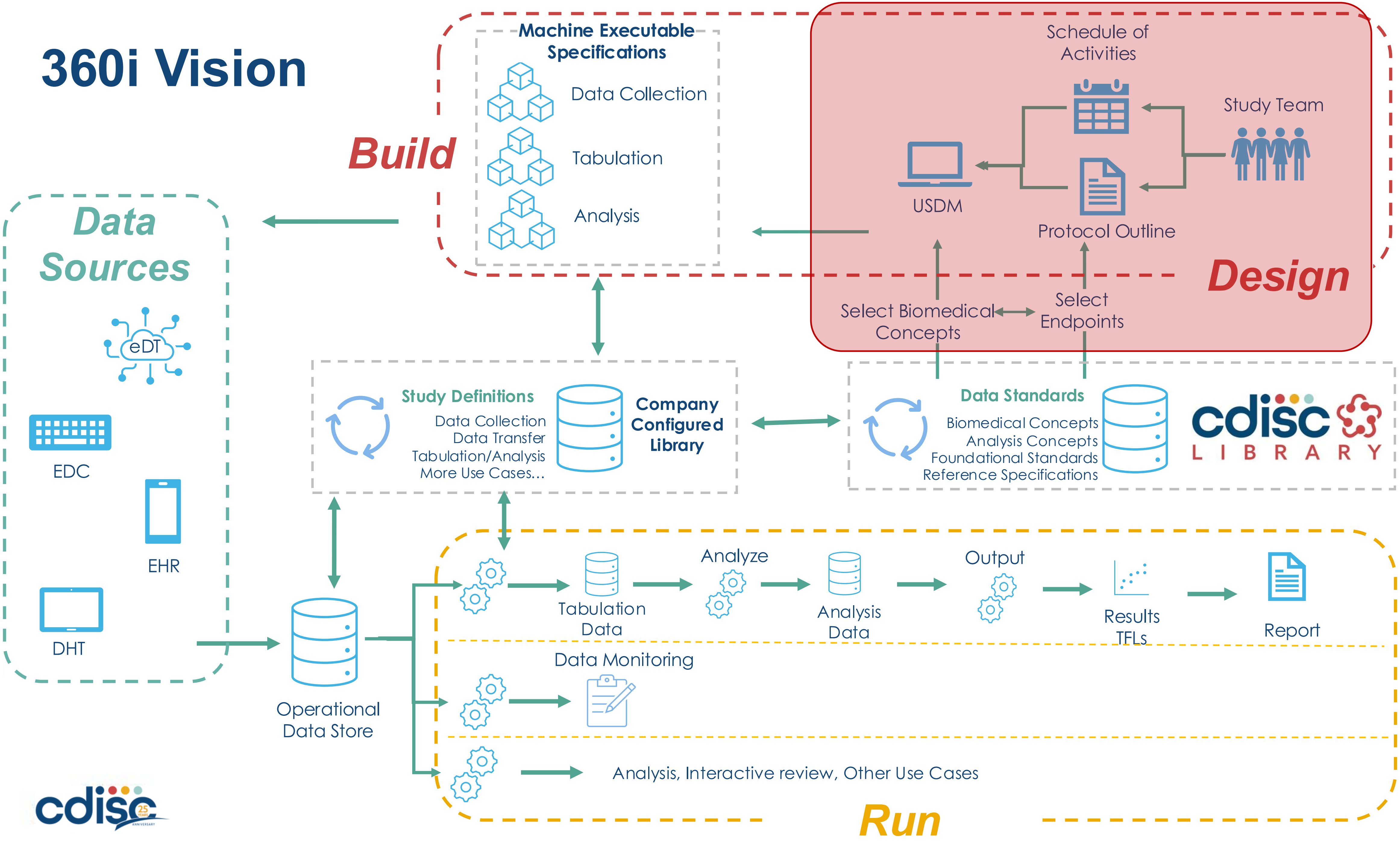
USDM & BCs serve as a catalyst for AI technologies by providing data with the full trial design context.

These study definitions, improved data quality, and enhanced cross-study consistency.

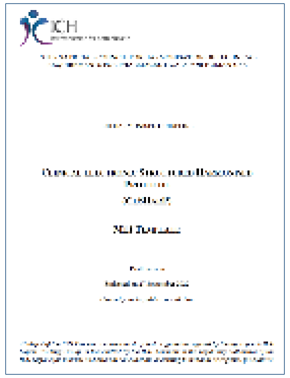
It addresses the critical limitation that "AI doesn't solve our context problems; it amplifies them" by bringing essential context to clinical trial data, empowering AI to operate more effectively.

What's Next?

360i Vision


Build
Machine Executable Specifications
Data Collection
Tabulation
Analysis
Schedule of Activities
Study Team
USDM
Protocol Outline
Select Biomedical Concepts
Select Endpoints
Design
Data Sources
eDT
EDC
EHR
DHT
Study Definitions
Data Collection
Data Transfer
Tabulation/Analysis
More Use Cases…
Company Configured Library
Data Standards
Biomedical Concepts
Analysis Concepts
Foundational Standards
Reference Specifications
cdisc LIBRARY
Operational Data Store
Tabulation Data
Analyze
Analysis Data
Output
Results TFLs
Report
Data Monitoring
Analysis, Interactive review, Other Use Cases
Run

USDM Phase 5



ICH & M11 Specifications
Align when final public version released

ICH harmonisation for better health

USDM Phase 5
Stability. The foundation for CDISC 360i. Scoping ongoing.

cdisc 360i

PROJECT PRISM
A Regulatory Cloud Collaborative Initiative

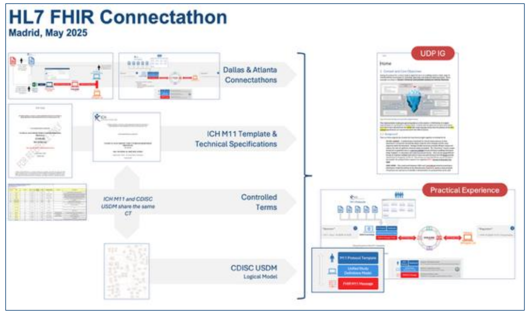
precisionFDA Regulatory Information Service Module
FDA-Industry Research Collaboration Agreement
(Public-Private Partnership)

FDA & PRISM
Working with FDA to pilot first electronic transfer of an M11 protocol as well as tooling to support

cdisc USDM
UNIFIED STUDY DEFINITIONS MODEL (DDF)

HL7 Vulcan & UDP
Working with HL7 Vulcan to build FHIR message to support exchange of USDM / M11 content.

VULCAN~UDP



EMA, M11 & CTIS
Working with EMA to align USDM with CTIS


EUROPEAN MEDICINES AGENCY

Home > Human regulatory: overview > Research and development > Clinical trials in human medicines > Clinical Trials Information System

Clinical Trials Information System

Share

The Clinical Trials Information System (CTIS) supports the flow of information between clinical trial sponsors, European Union (EU) Member States, European Economic Area (EEA) countries and the European Commission. CTIS went live with a searchable public website on 31 January 2022.

Human | Clinical trials

TransCelerate & Adoption
Sponsor and vendors working with USDM. Latest adoption will be visible at the TransCelerate 'Mission Possible' days in September

TransCelerate BIOPHARMA INC.

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Summary

- The electronic protocol (USDM v4) is with us now
- USDM is consistent and aligned with the ICH M11 Template
- USDM offers several routes to automation and efficiency gains
- Biomedical Concepts are key to achieving the full benefits

DIH Email

Scan Me

DIH LinkedIn

Scan Me

Mission Possible

Scan Me